A Topological Theory of Spatial OD Attribution in Traffic Networks

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ABSTRACT

The same origin–destination (OD) demand change can produce different cost accounts at different traffic boundaries. We establish when these accounts are interchangeable and how to compute and explain their differences on a fixed road network. Necessary and sufficient conditions depend jointly on network topology, the OD query, and the admissible region family. Arbitrary separating regions constrain weights within every cycle atom, whereas regions whose two sides are connected constrain only atoms on the queried block path; off-path weights can remain heterogeneous. For general marginal costs, one node-imbalance representation gives specified boundary accounts, exact unrestricted extrema, the best common-value approximation, and maximum boundary disagreement between models. A prefix scan also evaluates a prescribed finite family of straight screenlines exactly. Same-state traffic decompositions distinguish study and background responses under free-flow and congestion weights. In Sioux Falls, positive system contributions coexist with negative boundary accounts driven either by study congestion response or by background congestion response. Anaheim extends the spatial calculation and mechanism comparison to a larger network. An application to learned assignment models evaluates spatial explanation errors alongside flow errors. Together, the classification, computation and decomposition provide a framework for interpreting spatial OD contributions.

1. Introduction

Transportation sensitivity analysis asks how changes in origin–destination (OD) demand affect link flows and system outcomes (Tobin and Friesz, 1988; Josefsson and Patriksson, 2007; Patriksson and Rockafellar, 2003). For a differentiable assignment model or learned surrogate, integrated gradients allocate a finite outcome change across OD demands (Sundararajan et al., 2017). Resolving this allocation through individual link flows additionally locates the contribution on the network.

Consider a planner summarizing one demand change at several predetermined corridor cross-sections or regional boundaries, using a fixed road partition. The planner needs to know whether these summaries can replace one another; if they differ, the task is to locate the intervening weighted response and separate the queried traffic's adjustments from those of other traffic. The inputs are the fixed network, demand change, outcome function and candidate boundaries. The outputs are their OD accounts, an interchangeability criterion, and an additive explanation of their differences.

The network-wide contribution C_j, attribution on roads within a region, and signed boundary account $\phi_j(U)$ answer distinct questions. Different boundaries define different outcome functions, so equality is a structural property to establish rather than a prerequisite for attribution. The proposed classification determines when the summaries share a common boundary value across conservative responses. When they do not, a node representation computes their differences and identifies the traffic responses responsible. A negative boundary account can therefore be a meaningful local cost response even when the network-wide OD contribution is positive.

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The question differs from tracking physical OD vehicles through screenlines or cordons (Sheffi, 1985; van Zuylen and Willumsen, 1980; Bar-Gera, 2002): an OD-indexed sensitivity describes the network-wide response, including other ODs' route changes, and weighting it by marginal travel times or costs preserves additive completeness while creating interior attribution sources and sinks — on two series links, a conservative unit response $(1, 1)$ with weights (w_1, w_2) balances at the intermediate node only if $w_1 = w_2$. Vehicle conservation alone thus does not make outcome contributions interchangeable across boundaries.

This distinction matters for explanations of conservative traffic models: physical penalties, output projections, and exactly conservative parameterizations constrain the predicted flows (Raissi et al., 2019; Shi et al., 2021; Valente et al., 2025; Richter-Powell et al., 2022), but conservation of those flows does not by itself make a learned model's attribution interchangeable across boundaries. Exact conservation transfers to flow-valued integrated gradients; outcome weighting then imposes additional conditions on the network and the outcome definition; and approximate output conservation may fail at the first transfer step, because a small residual need not have small OD derivatives.

The contribution has three connected parts. First, necessary and sufficient topological conditions characterize boundary interchangeability for both arbitrary separating regions and regions with two connected sides. The region family changes the answer: only queried block paths constrain weights in the latter case. Second, a common node representation computes selected accounts, unrestricted extrema, approximate interchangeability and exact model disagreement; a reusable geometric scan handles prescribed straight screenlines. Third, same-state decompositions explain spatial differences through study and background route responses under free-flow and congestion weights. The cycle and block tools are classical; their coupling to the OD query, region family and fixed attribution object is the central new result. Endpoint identities and derivative-residual bounds support the calculation. Learning comparisons demonstrate how the spatial framework evaluates assignment explanations.

2. Related work

Attribution and cost allocation. Our edge-level object is input-specific conductance (Dhamdhere et al., 2019; Lundstrom et al., 2022): demand is the input, link flow is an intermediate variable, and the traffic outcome is the output. Its sum over links gives the integrated-gradient contribution of an OD. Continuous marginal cost allocation also underlies Aumann–Shapley sharing (Aumann and Shapley, 1974; Billera et al., 1978). Those scalar allocations and computational-graph completeness do not impose node balance on a traffic network. We analyze that additional property.

Conserved attribution flows on graphs. Graph-based explanation methods construct attribution flows with boundary-consistency or conservation properties on directed acyclic graphs: Shapley Flow makes edge attributions consistent across different choices of explanatory boundary, with conservation at internal nodes following from the efficiency axiom together with boundary consistency, and the recursive Shapley value recursively constructs edge-attribution flows that conserve at internal nodes (Wang et al., 2021; Singal et al., 2021). Those works design new attribution rules satisfying such axioms on computational or causal graphs. Here the graph is a physical traffic network with cycles, the edge-level conductance is fixed by the assignment model and outcome, and the question is when these existing outcome-weighted sensitivities have equal signed sums across OD-separating cuts, uniformly over conservative responses, rather than when a newly constructed rule is conserved.

Traffic models and sensitivity. Equilibrium sensitivity analysis studies demand-to-flow derivatives and their aggregation (Tobin and Friesz, 1988; Yang and Bell, 2007; Josefsson and Patriksson, 2007; Patriksson and Rockafellar, 2003). Learned assignment maps, implicit optimization layers, and logit equilibrium metamodels make such derivatives available to explanations (Liu et al., 2023; Amos and Kolter, 2017; Lassen et al., 2025). Physical OD tracking through assignment and cordon counts concerns commodity flows (Sheffi, 1985; van Zuylen and Willumsen, 1980; Bar-Gera, 2002); it does not by itself establish transferable budgets for marginal-cost-weighted attributions.

Conservation and graph structure. Physics-informed losses and conservative architectures constrain physical outputs (Raissi et al., 2019; Shi et al., 2021; Richter-Powell et al., 2022); projections can enforce

constraints exactly (Valente et al., 2025; Baez et al., 2024). We examine their transfer to flow and outcome attributions; for approximate conservation, derivative bounds of Landau–Kolmogorov type are classical (Shadrin, 2014), and Appendix A.4 transfers such control to attribution residuals. The relevant circulation space and its decomposition into circuit classes are classical (Oxley, 2011; Hopcroft and Tarjan, 1973). The new classification couples this decomposition to queried OD directions and their transitive equality constraints.

3. Attribution objects and balance criteria

3.1. Edge attribution and the accounting question

Let $\mathcal{G} = (V, E)$ be connected, with incidence matrix $B \in \{-1, 0, 1\}^{n\times m}$ (negative at a link's tail), OD vectors $\mu_j = e_{t_j} - e_{s_j}$, $s_j \neq t_j$, and $M = [\mu_1, \dots, \mu_K]$. Reciprocal and parallel arcs remain distinct edges when orientations are forgotten. Let $f : \Omega \subset \mathbb{R}^K \to \mathbb{R}^m$ map demands to flows on an open demand domain Ω, and $T : \mathbb{R}^m \to \mathbb{R}$ be a specified edgewise outcome. Assume both maps are C^1 on open neighborhoods of the relevant paths. With $\Delta q = q^1 - q^0$ and $\gamma(\lambda) = q^0 + \lambda\Delta q$, define

$$G_{:,j} = \Delta q_j \int_0^1 J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda, \tag{1}$$

$$\widetilde{C}_{:,j} = \Delta q_j \int_0^1 D_\lambda J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda, \quad D_\lambda = \operatorname{diag}(\nabla T(f(\gamma(\lambda)))), \tag{2}$$

$$C_j = \mathbf{1}^\top \widetilde{C}_{:,j}, \qquad \sum_j C_j = T(f(q^1)) - T(f(q^0)). \tag{3}$$

G attributes flow changes; $\widetilde{C}$ resolves the outcome contribution through links; C_j is its scalar OD contribution. Their columns are signed system sensitivities, including other ODs' route adjustments, rather than physical commodity flows. Entry $\widetilde{C}_{e,j}$ is input-specific conductance through link e (Dhamdhere et al., 2019).

If $Bf(q) = Mq$, differentiation and integration give

$$BJ_f(q) = M, \qquad BG_{:,j} = \mu_j \Delta q_j. \tag{4}$$

The same conclusion holds for $Bf(q) = Mq + b$ with constant background demand balance b. Outcome attribution is *same-OD balanced* when $B\widetilde{C}_{:,j} = \beta_j \mu_j$ for some $\beta_j \in \mathbb{R}$. The scale β_j is a boundary value, generally different from C_j: on two series links with unit weights, $C_j = 2\Delta q_j$ but $\beta_j = \Delta q_j$.

For $s_j \in U \subseteq V \setminus \{t_j\}$, orient the cut account as $\phi_j(U) = -\mathbf{1}_U^\top B\widetilde{C}_{:,j}$, and write $b_j = B\widetilde{C}_{:,j}$. Any two separating sets satisfy

$$\phi_j(U') - \phi_j(U) = -\sum_{v\in U'\setminus U} b_{jv} + \sum_{v\in U\setminus U'} b_{jv}; \tag{5}$$

for nested sets $U \subseteq U'$ the second term vanishes, leaving $\phi_j(U') - \phi_j(U) = -\sum_{v\in U'\setminus U}(B\widetilde{C}_{:,j})_v$. The difference between two boundaries is thus the enclosed imbalance, with incoming and outgoing regions entering with opposite signs. For a separable outcome $T(F) = \sum_e g_e(F_e)$, the account itself also has a direct reading. With $\delta^+(U)$ and $\delta^-(U)$ the arcs leaving and entering U,

$$H_U(F) = \sum_{e\in\delta^+(U)} g_e(F_e) - \sum_{e\in\delta^-(U)} g_e(F_e), \qquad \phi_j(U) = \mathrm{IG}_j[\, H_U \circ f \,], \tag{6}$$

because each entry $\widetilde{C}_{e,j}$ of Eq. (2) is the integrated gradient of the scalar outcome $g_e \circ f$, and the incidence signs turn $-\mathbf{1}_U^\top B\widetilde{C}_{:,j}$ into the signed boundary sum of H_U. For TSTT, $\phi_j(U)$ therefore attributes the demand change to the excess of the total travel cost on the region's outgoing arcs over that on its incoming arcs: a boundary-related outcome in its own right, neither the region's internal cost nor a vehicle count.

Proposition (cut-consistency equivalence)**.** *Same-OD balance holds exactly when every separating cut has one common signed value* $\phi_j(U) = \beta_j$.

Table 1
Three spatial summaries of one OD attribution. E_R is a road set specified before aggregation.

Object	Transportation question
$C_j = \sum_e \widetilde{C}_{e,j}$	How much does this OD contribute to the system outcome change?
$\sum_{e\in E_R} \widetilde{C}_{e,j}$	How much of that contribution is located on the specified regional roads?
$\phi_j(U)$	What is its signed aggregation on the specified boundary?

Indeed, balance implies this identity. Conversely, comparing $\{s_j\}$ with $\{s_j, v\}$ forces zero divergence at every interior node; the two endpoint entries are then opposite because incidence columns sum to zero.

Throughout, $\mathcal{U}_j^{\text{all}}$ contains all sets separating s_j from t_j; $\mathcal{U}_j^{\text{conn}}$ restricts both induced sides to be connected in the underlying undirected network; and $\mathcal{U}_j^{\text{geo}}$ is a specified finite family of straight screenline regions, filtered for two-sided connectivity. The unrestricted extrema below are closed-form exact values. Sioux Falls connected extrema use complete enumeration. Geometric extrema and Anaheim's selected connected-region ranges are exact only over their stated finite families, and do not determine extrema over all connected regions.

3.2. Pointwise guarantees and realized balance

At fixed weights $D = \operatorname{diag}(w)$, realized-model balance requires $BDJ_j = \alpha_j \mu_j$ for the model's actual response. Universal directional balance requires $BDx \in \operatorname{span}\{\mu_j\}$ for every $Bx = \mu_j$, allowing its coefficient to depend on x. *Structural compatibility* additionally requires one common coefficient across all such responses. The last property is what we classify. For non-collinear queries with shared weights, universal directional balance already implies structural compatibility because $BD \ker B \subseteq \bigcap_j \operatorname{span}\{\mu_j\} = \{0\}$; a query and its reverse alone do not suffice. Pointwise balance along the demand path implies integrated balance; the converse can fail through cancellation. Appendix A.3 gives examples and restricted-response results. Thus a realized attribution can balance without satisfying the universal structural conditions.

4. Topology, OD queries and admissible regions

Take the block-cut tree of the underlying undirected multigraph. Its non-bridge blocks, together with each bridge as a singleton, are the *cycle atoms*. They are the classical graphic-matroid circuit classes, with each bridge retained separately. The *OD block path* $\mathcal{P}(s,t)$ is the atom sequence on the unique block-cut-tree path between the OD endpoints. For several queries, merge all atoms traversed by each $\mathcal{P}_j = \mathcal{P}(s_j, t_j)$ and take the transitive closure. The resulting edge sets $\bar{E}_1, \ldots, \bar{E}_{K'}$ are the *OD-compatibility classes*; atoms on no queried path remain separate.

Theorem 1 (single-OD classification, static form)**.** *Let $\mathcal{G}$ be a connected multigraph with one OD pair (s,t), $s \neq t$, let $w \in \mathbb{R}^m$, and let $D = \operatorname{diag}(w)$. The* response-independent same-OD balance property

$$\exists\, \alpha \in \mathbb{R} : \qquad BDx = \alpha\, \mu \quad \textit{for every } x \in \mathbb{R}^m \textit{ with } Bx = \mu, \tag{7}$$

with the scale α independent of the response x, holds ***if and only if***

(i) *w is constant within every non-bridge atom; and*
(ii) *all atoms on the s–t block path $\mathcal{P}(s,t)$ share one common value.*

In that case every conservative response induces the same boundary vector: $BDx = w^*\mu$ for every x with $Bx = \mu$, where $w^* := w|_{\bigcup \mathcal{P}(s,t)}$ is the common value of (ii) and $\alpha = w^*$ is the scale in (7).

Condition (i) concerns every atom, including those off the OD path. Their constants may differ; only the traversed atoms must share the scale w^*. The proof adds a virtual edge from t back to s (Appendix A.2): conservative OD responses then become circulations on the extended graph, so the boundary vector is response-independent exactly when diagonal weighting preserves the circulation space — equivalently, when every weighted cycle has zero node divergence — which forces equal weights within each cycle atom and one shared weight along the atoms the OD block path connects.

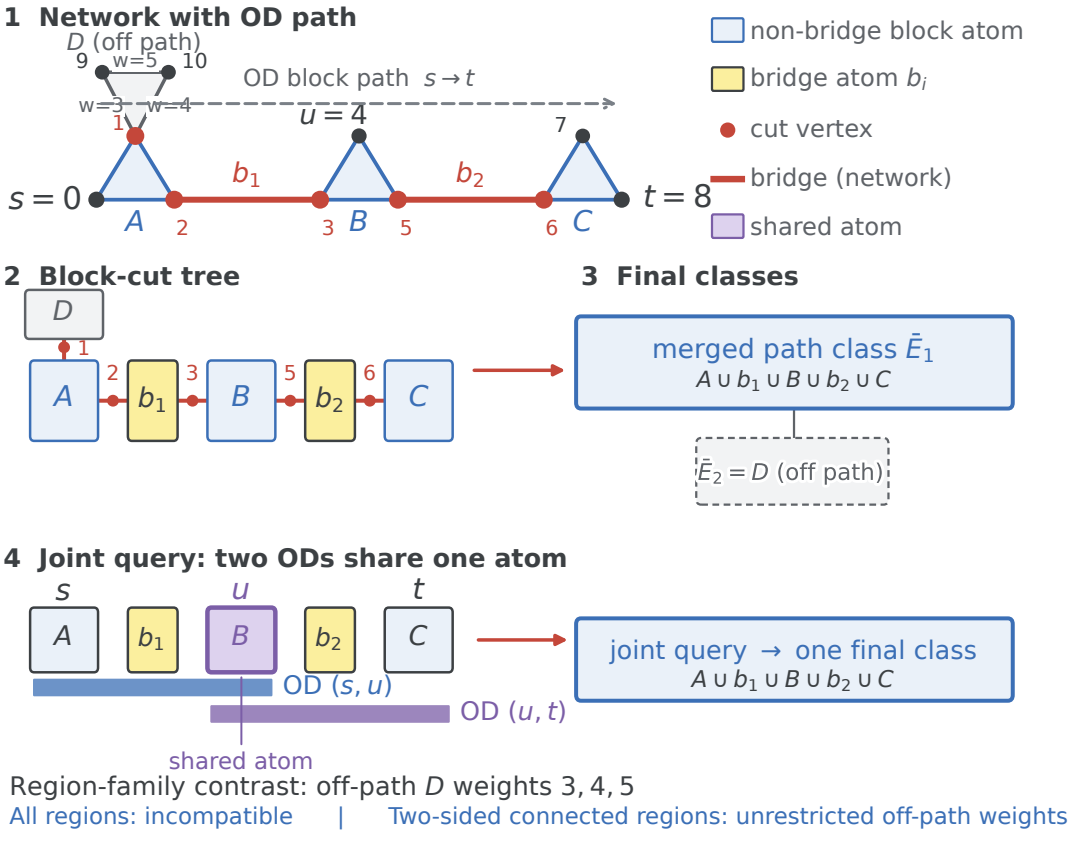


Figure 1: OD queries merge the atoms they traverse. Two paths sharing atom B merge the full chain transitively; off-path atom D stays separate. For the connected-region comparison, atom D has heterogeneous weights $3, 4, 5$: these are invisible to the queried connected boundaries, but violate unrestricted interchangeability. Orientations are omitted; node numbering starts at zero.

Theorem 2 (multi-OD classification, static form at a shared realized flow state)**.** *Let $\mathcal{G}$ be connected, let each OD pair satisfy $s_j \neq t_j$, and let $w \in \mathbb{R}^m$, $D = \mathrm{diag}(w)$. The response-independent same-OD balance property (7) — for each queried j, some $\alpha_j \in \mathbb{R}$ with $BDx = \alpha_j \mu_j$ for every x with $Bx = \mu_j$ — holds simultaneously for the queried ODs* ***if and only if*** *w is constant on each OD-compatibility class $\bar{E}_k$.*

The classification concerns all signed responses compatible with vehicle conservation; an equilibrium model generally realizes only a subset of these responses (Appendix A.3). All queries use the same realized weight vector. Queries with $\Delta q_j = 0$ have zero attribution columns and need not enter the closure. Requiring all ordered pairs of distinct vertices collapses the classes to one per connected component (Corollary A.4).

Corollary 3 (path form: endpoint implication)**.** *Let $f \in C^1$ be conservative on an open set, $T \in C^1$, $\gamma(\lambda) = q^0 + \lambda \Delta q$ the straight-line path with $\Delta q \neq 0$, and $w(\lambda) = \nabla T(f(\gamma(\lambda)))$, $D_\lambda = \mathrm{diag}(w(\lambda))$, shared by all participating ODs. If the equivalent conditions of Theorems 1 and 2 hold at $w(\lambda)$ for every $\lambda \in [0, 1]$, then, for every participating j simultaneously, the endpoint balance*

$$B\widetilde{C}_{:,j} = \beta_j \, \mu_j$$

holds with

$$\beta_j = \Delta q_j \int_0^1 w^*_{k(j)}(\lambda) \, d\lambda,$$

where $w^*_{k(j)}$ is the common value on the compatibility class $k(j)$ containing the atoms of $\mathcal{P}_j$.

Table 2
Constructive query-dependent compatibility on Figure 1. Ranges use all 512 separating sets per query and a conservative unit response; the weighted circulation residual is below 5×10^{-16} in all stages, so these ranges are independent of that response.

Stage	$w_{A,b_1,B}$	$w_{b_2,C}$	w_D	Queried ODs	Account ranges
Initial query	1	2	3	(s,u)	$[1,1]$
Add query	1	2	3	$(s,u);(u,t)$	$[1,1];[1,2]$
Adjust merged class	1	1	3	$(s,u);(u,t)$	$[1,1];[1,1]$

Figure 1 becomes a constructive design example in Table 2. We use its 11-node, 14-edge graph with $s = 0$, $u = 4$, $t = 8$ and the linear structural outcome $T_w(F) = w^\top F$. Initially the query (s,u) joins A, b_1, B, while b_2, C, and D can retain their own constants. Adding (u,t) joins the remaining chain through their shared atom B. Adjusting only b_2 and C then restores a common account for both queries while retaining the unrelated atom's weight 3. The calculation uses the minimum-norm solution of $Bx = \mu$ and checks $BDZ = 0$ on a basis of $\ker B$, which establishes response independence of the displayed boundary vector. The theorem supplies the uniform common-account guarantee in stages one and three. These intentionally chosen linear weights illustrate compatibility; they are not asserted to be the marginal costs of the original TSTT outcome.

Off-path atoms are constrained here because a signed circulation can be added to any conservative response, and heterogeneous weights can turn it into node divergence. Connected regions change this conclusion.

4.1. Compatibility for connected traffic regions

The admissible region family changes the structural answer. Let

$$\mathcal{U}_{st}^{\mathrm{conn}} = \{U \subset V : s \in U,\ t \notin U,\ G[U] \text{ and } G[V \setminus U] \text{ are connected}\},$$

where connectivity uses the underlying undirected multigraph. These are separating bonds: both sides of the boundary form connected regions.

Theorem 4 (two-sided connected-region classification)**.** *For a connected loopless multigraph, $\mu = e_t - e_s$, and $D = \mathrm{diag}(w)$, the property*

$$\exists \alpha \in \mathbb{R} \quad \forall x \text{ with } Bx = \mu \quad \forall U \in \mathcal{U}_{st}^{\mathrm{conn}} : -\mathbf{1}_U^\top BDx = \alpha \tag{8}$$

holds if and only if every edge in the atoms of $\mathcal{P}(s,t)$ has one common weight α. Weights of edges outside those atoms are arbitrary, even within an off-path cycle atom.

The proof in Appendix A.6 uses connected prefix and suffix regions within each traversed atom. Each admissible boundary crosses edges in just one such atom, which explains why off-path heterogeneity is invisible to this region family. This is a universal fixed-weight statement; agreement for one realized response need not imply its conditions.

Corollary 5 (several queries on connected regions)**.** *At one shared weight vector, Eq. (8) holds for all queried ODs if and only if edges in each group of atoms joined by the queried block paths have one common weight. An atom on no queried path has no weight restriction, including no within-atom restriction.*

Thus restricting attention to connected traffic regions permits additional outcome heterogeneity. It changes the compatibility classification, while the unrestricted extrema and error formulas below retain their own stated boundary family.

A constructive distinction on Figure 1. Keep weights 1 on A, b_1, B, weights 2 on b_2, C, but set the three off-path D weights to $3, 4, 5$. The first query (s,u) admits common account 1 on every two-sided connected region for every conservative response. The second query (u,t) initially has connected accounts spanning $[1,2]$ and fails that criterion. Changing only b_2, C to weight 1 makes both queries compatible on connected

regions, while D remains internally heterogeneous. For each query there are 512 unrestricted separating sets but only 5 two-sided connected sets. Checking a particular response and a complete circulation basis confirms the universal connected-region assertions; this is stronger than checking sampled responses. A unit signed circulation in D gives unrestricted accounts $[-1, 3]$ after the merge, while all connected accounts remain 1. Thus the unrestricted criterion still fails. This fixed structural outcome illustrates the theorem; its weights do not replace the original TSTT marginal costs. The reproducible record is provided with the code.

4.2. Linear outcomes and transportation costs

Corollary 6 (linear outcome functionals)**.** *Let $f \in C^1$ be exactly conservative on an open neighborhood of the attribution path, and let $T(F) = a^\top F$ be a linear functional of realized flows. Then $\nabla T \equiv a$, so response-independent cut balance holds if and only if a is constant on the compatibility classes — a condition on the outcome definition alone, checkable by inspection before any model is trained. The pointwise classification then holds at every λ, with common cut account $\beta_j = \Delta q_j\, a|_{\bigcup \mathcal{P}_j}$ exactly.*

Nonlinear compatible outcomes also exist: if $T(F) = \Phi(\sum_{e\in\bar{E}_1} F_e, \ldots, \sum_{e\in\bar{E}_{K'}} F_e)$, its gradient is class-constant at every flow. More familiar traffic outcomes require equality of their actual marginal weights. The Beckmann potential has weights $t_e(F_e)$; total system travel time (TSTT) has weights $t_e(F_e) + F_e t'_e(F_e)$; a separable index $\sum_e g_e(F_e)$ has weights $g'_e(F_e)$. Heterogeneous link parameters and congestion typically violate these equalities. A single-link outcome has a nonzero common boundary value only when the queried OD block path is precisely that singleton bridge.

4.3. Endpoint cancellation and network representation

On two series links, an outcome whose marginal weights oscillate around unity can integrate to the same balanced column $(1, 1)$ as the linear outcome on one demand segment, yet fail to balance on another; the linear outcome gives the same composite demand-to-outcome map with balanced edge attributions on every segment (Example A.7). The specified edgewise outcome therefore matters even at fixed network topology.

The network representation matters as well: subdividing a link leaves the composite map unchanged but creates interior divergence $\tau(2\alpha - 1)\Delta q$, and even the balanced split $\alpha = 1/2$ halves the original cut value (Example A.8). Response independence concerns conservative responses on a fixed network; it does not make the account invariant to road subdivision.

5. Exact spatial computation

Once the classes are constructed, structural compatibility is tested by

$$\eta(\lambda) = \max_k \max_{e\in\bar{E}_k} |w_e(\lambda) - \bar{w}_k(\lambda)|, \qquad \bar{w}_k = |\bar{E}_k|^{-1} \sum_{e\in\bar{E}_k} w_e.$$

The joint condition is exactly $\eta = 0$. Class construction uses the block-cut tree and transitive merging; testing a weight vector then needs one edge scan. No fitted-model derivatives are required for a linear outcome.

For a realized integrated attribution, define

$$d_j = (I - P_{\mu_j}) B\widetilde{C}_{:,j}, \qquad P_{\mu_j} = \frac{\mu_j \mu_j^\top}{\|\mu_j\|_2^2}. \tag{9}$$

Endpoint cut consistency holds precisely when $d_j = 0$; only then does $\mu_j^\top B\widetilde{C}_{:,j}/\|\mu_j\|_2^2$ represent a common cut value, whatever mechanism produced the balance (Section 3.2).

Without a connectivity restriction, each interior node can enter the separating set independently. Therefore Eq. (5) gives

$$S_j := \max_U \phi_j(U) - \min_U \phi_j(U) = \sum_{v\notin\{s_j,t_j\}} |(B\widetilde{C}_{:,j})_v|. \tag{10}$$

The extrema themselves have a closed form: writing $b_j = B\widetilde{C}_{:,j}$, $(x)_+ = \max(x, 0)$, and summing over interior nodes $v \notin \{s_j, t_j\}$,

$$\phi_j^{\min} = \phi_j(\{s_j\}) - \sum_v (b_{jv})_+, \qquad \phi_j^{\max} = \phi_j(\{s_j\}) + \sum_v (-b_{jv})_+, \tag{11}$$

since the minimizing set retains exactly the interior nodes with positive divergence. Subtracting gives, for every separating set U with $I_j = V \setminus \{s_j, t_j\}$,

$$\phi_j(U) - \phi_j^{\min} = \sum_{v \in I_j \setminus U} (b_{jv})_+ + \sum_{v \in I_j \cap U} (-b_{jv})_+, \tag{12}$$

so the gap to the unrestricted minimum is exactly the positive divergence the region omits plus the negative divergence it contains. When $\phi_j(\{s_j\}) > 0$, the unrestricted family therefore contains a negative account whenever the interior positive divergence $\sum_v (b_{jv})_+$ exceeds the origin-cut account: node imbalance, extremal location, and sign crossing are the same statement. $S_j = 0$ is equivalent to $d_j = 0$. The projector in Eq. (9) changes only the two endpoint entries. Cauchy–Schwarz on the $n - 2$ interior entries therefore gives

$$S_j = \sum_{v \notin \{s_j, t_j\}} |(d_j)_v| \le \sqrt{n-2}\, \|d_j\|_2. \tag{13}$$

For network regions whose induced subgraphs and complements are both connected, define S_j^{conn} by restricting the same extrema to that family. Then $S_j^{\text{conn}} \le S_j$; a small restricted range alone does not establish the full structural criterion. We report both ranges as percentages of the absolute origin-cut value $|\phi_j(\{s_j\})|$, which remains explicit because the ratio depends on that denominator (absolute ranges when it is zero).

Corollary 7 (best common boundary value)**.** *For the unrestricted family, the unique best constant in maximum absolute account error is*

$$\beta_j^\star = \frac{\phi_j^{\max} + \phi_j^{\min}}{2} = \frac{(b_j)_{t_j} - (b_j)_{s_j}}{2}, \qquad \min_{\beta \in \mathbb{R}} \max_{U \in \mathcal{U}_j^{\text{all}}} |\phi_j(U) - \beta| = \frac{S_j}{2}.$$

Hence all separating accounts lie within a prescribed absolute tolerance ε of some common value if and only if $S_j \le 2\varepsilon$.

Proof. For any constant, its largest error at the two attained extrema is at least half their difference; their midpoint attains this bound for every account. Equation (11) and $\mathbf{1}^\top b_j = 0$ give the midpoint formula. □

The coefficient agrees with the endpoint projection in Eq. (9). It approximates the original accounts; it does not change their edge attribution or define a new system contribution.

5.1. Exact boundary error and response supervision

For a query j and any edge vector h, define the seminorm

$$\mathcal{K}_j(h) = \frac{1}{2}\left(\left| (Bh)_{t_j} - (Bh)_{s_j} \right| + \sum_{v \notin \{s_j, t_j\}} |(Bh)_v| \right). \tag{14}$$

Corollary 8 (exact maximum boundary error)**.** *Let $e_j = \widetilde{C}^L_{:,j} - \widetilde{C}^S_{:,j}$ be the difference between learned and reference attribution columns. Then*

$$\max_{U : s_j \in U,\ t_j \notin U} |\phi_j^L(U) - \phi_j^S(U)| = \mathcal{K}_j(e_j). \tag{15}$$

At a common demand q, write $h_j(q) = D^L(q) J_j^L(q) - D^S(q) J_j^S(q)$. For any subfamily $\mathcal{U}_j$ of separating boundaries,

$$\epsilon_j^{\mathcal{U}} := \max_{U \in \mathcal{U}_j} |\phi_j^L(U) - \phi_j^S(U)| \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda. \tag{16}$$

Algorithm 1: Spatial accounts from an attribution column

Input: incidence B, query (s_j, t_j), edge attribution c_j, selected regions U, W, and optionally a comparison column c'_j.

1. Compute $b_j = Bc_j$; return its node distribution.
2. Return $\phi_j(U) = -\mathbf{1}_U^\top b_j$ and $\phi_j(U) - \phi_j(W) = -(\mathbf{1}_U - \mathbf{1}_W)^\top b_j$.
3. Scan interior nodes to return the extrema in Eq. (11) and their sets $\{s_j\} \cup \{v : b_{jv} > 0\}$ and $\{s_j\} \cup \{v : b_{jv} < 0\}$.
4. If c'_j is supplied, return $\mathcal{K}_j(c_j - c'_j)$ as the exact maximum absolute error over unrestricted separating regions.

The incidence product and scan cost $O(|E| + |V|)$ per column; selected-region sums cost $O(|V|)$. The same node representation serves all these tasks. The columns must already have been computed; their reference-attribution cost remains separate.

Table 3
Same-task spatial aggregation after reference columns are available. Median batch times; reference generation is excluded. The 120-column workload repeats the six columns twenty times and measures repeated-query cost, not additional empirical evidence.

Columns	Accounts	Prefix scan (ms)	Direct crossings (ms)
6	1642	4.16	28.51
120	32840	87.92	1176.15

The proof in Appendix A.5 applies Eq. (11) to the error column. Evaluation requires one incidence multiplication and node scan, without enumerating boundaries. Connected regions and straight screenlines inherit an upper bound; their restricted maxima need not equal $\mathcal{K}_j(e_j)$. The endpoint term must be retained: the two explanations may each be cut-consistent but have different common boundary values, which internal imbalance alone misses.

The seminorm evaluates spatial explanation errors directly. It also defines reference-matching losses, whose complete specifications are collected in Appendix B.7. Their comparison is an application of the spatial framework (Section 7).

5.2. Exact scanning of predefined straight screenlines

Fix D directions before evaluating attribution. For each direction, sort nodes by their coordinate projection and group equal projection values; no screenline splits such a group. A candidate position is a gap between adjacent groups. Forward and reverse union–find sweeps record whether each prefix and suffix induces a connected subgraph. This geometry and topology preprocessing is reusable across queries, demand paths and models. For one attribution column, form $b_j = B\widetilde{C}_{:,j}$ and prefix sums $P_{j,k} = \sum_{i \le k} (b_j)_{\pi(i)}$. If s_j lies in the prefix and t_j outside, its account is $-P_{j,k}$; for the reverse orientation it is $P_{j,k}$ because $\mathbf{1}^\top b_j = 0$. Retain only endpoint-separating gaps with both sides connected. Scanning these values gives exact accounts and extrema over this prescribed finite screenline family. It does not optimize over every possible direction in the plane. Preprocessing takes $O(D(|V| \log |V| + |E|\alpha(|V|)))$; after preprocessing, a column takes $O(|E| + D|V|)$ using sparse incidence multiplication. Duplicate regions may be removed without changing extrema.

On Sioux Falls, 36 directions at five-degree intervals give 1642 admissible query–position accounts across six queries (duplicates across directions retained for both algorithms). Direct summation recomputes outgoing minus incoming crossing-edge contributions on exactly these same oriented regions. The largest account discrepancy was 2.2e-11 TSTT units. Reusable preprocessing took 83.48 ms.

Scope and approximate conservation. The classification fixes the road representation, outcome weights and admissible regions, and quantifies over all signed conservative responses. For a model with residual $r(q) = Bf(q) - Mq - b$ and constant b, the corresponding flow-attribution residual is

$$R_j := BG_{:,j} - \mu_j \Delta q_j = \Delta q_j \int_0^1 \partial_j r(\gamma(\lambda))\, d\lambda, \qquad \sum_j R_j = r(q^1) - r(q^0). \tag{17}$$

Small endpoint residuals control the sum, but individual columns can cancel. Appendix A.4 gives derivative-regularity conditions, a local bound, sharpness examples and an extension to weighted accounts. The traffic experiments below use exactly conservative models.

6. Numerical evidence and transportation interpretation

For Sioux Falls, all reported absolute TSTT accounts use benchmark flows in veh/h and time units of 0.01 h; multiply a reported number by 0.01 to obtain the hourly vehicle-hour total. This convention applies to the Sioux Falls attribution and error tables. Anaheim instead uses minutes, as specified separately in Section 6.6.

6.1. Study design and local mechanism

The Sioux Falls experiment compares an exactly conservative learned surrogate with direct fixed-path logit-SUE on eight OD queries. The two evidence types divide the labour: Figure 1 illustrates the classification's query dependence on a network whose queries merge atoms differently, whereas here all 76 arcs lie in one compatibility class, the structural weight condition is the same for every OD, and the experiment instead measures what boundary differences heterogeneous realized cost weights produce under equilibrium responses. The OD differences reported below thus arise from realized responses, weight dispersion, and integration, not from different compatibility classes. Each study OD uses ten fixed paths. We compare zero background (A), frozen background flows (B), and joint re-equilibration at fixed background demands (C). Setting C loads all 528 positive-demand OD pairs but attributes changes only to the eight study demands, selected as the eight largest published demands — all involving node 10, a local rather than representative comparison. Appendix B gives the full settings.

Each study demand moves from $0.2q_{\text{base}}$ to $1.6q_{\text{base}}$ along a straight line (q_{base}: the eight published demands, 3,900–4,400 veh/h); the eight-query experiments use this demand segment unless otherwise stated. The three settings differ only in how the remaining demands are handled.

The settings also differ in the redistribution directions their fixed path sets can express: setting A's path-exchange space T_{swap} (Appendix A.3) has dimension 32, a proper subspace of the 53-dimensional circulation space, while setting C's full-background path set spans the complete space ($53 = m - n + 1$; Appendix B.1). Spanning the circulation space is a statement about linear directions, not about which signed responses the logit-SUE Jacobian realizes.

Uniform weights are a compatible control. The inactive-link outcome $T(F) = F_{1\to2}$ gives zero attribution because that arc lies outside the setting-A surrogate path union (setting C's path sets cover every arc). The active-link outcome $T(F) = F_{10\to16}$ gives $BDx = \mu_{(10,16)}x_{10\to16}$: direction is preserved for that OD, while circulations can change the scale. TSTT instead weights the response by heterogeneous marginal social costs and produces interior node imbalance. Figure 2 shows these three patterns on the same realized learned response. For the seed-0 TSTT account, adding node 17 to $U = \{10\}$ lowers $\phi_j(U)$ from 5.77×10^4 to 4.34×10^4 — the largest of the 22 single-node enlargements, accounting for about 92% of the unrestricted range, with the direct-SUE counterpart at 24.80% (Section 6.3); both displayed regions and their complements are network-connected. The free-flow component contributes 84.7% of the node-17 imbalance, the congestion premium 15.3%; the ordering reverses at the negative minima of Section 6.4, where the congestion premium dominates (Table 6). All incident links have free-flow time 2 except $10 \leftrightarrow 17$, which has time 8; exact vehicle balance gives

$$[B\,\mathrm{diag}(t_0)G_j]_{17} = (8-2)(G_{10\to17,j} - G_{17\to10,j}).$$

The exchange cancels in vehicle units but not under heterogeneous costs.

6.2. Background traffic changes the same OD queries

Table 4 combines the three demand settings. With zero background, learned and direct-SUE ranges separate into four large and four small values. This reflects the realized demand response rather than fixed OD characteristics; Appendix C.2 relates the small ranges to small pointwise imbalance.

Freezing background link flows changes congestion while suppressing the background route response; letting routes re-equilibrate produces large ranges for all eight queries — $(10, 16)$ moves from 26.94% in A to 109.04% in C, the initially near-balanced $(10, 11)$ from 0.0075% to 84.67% (percentages on each setting's own origin-cut value; per-query absolute denominators in Appendix C.2).

Under re-routing, connected boundaries retain 60.41–76.85% of the unrestricted range (connected ranges 31.57–127.72% of the origin-cut value): connectivity materially narrows the background envelope but does not restore cut consistency, and the two families should not be interchanged.

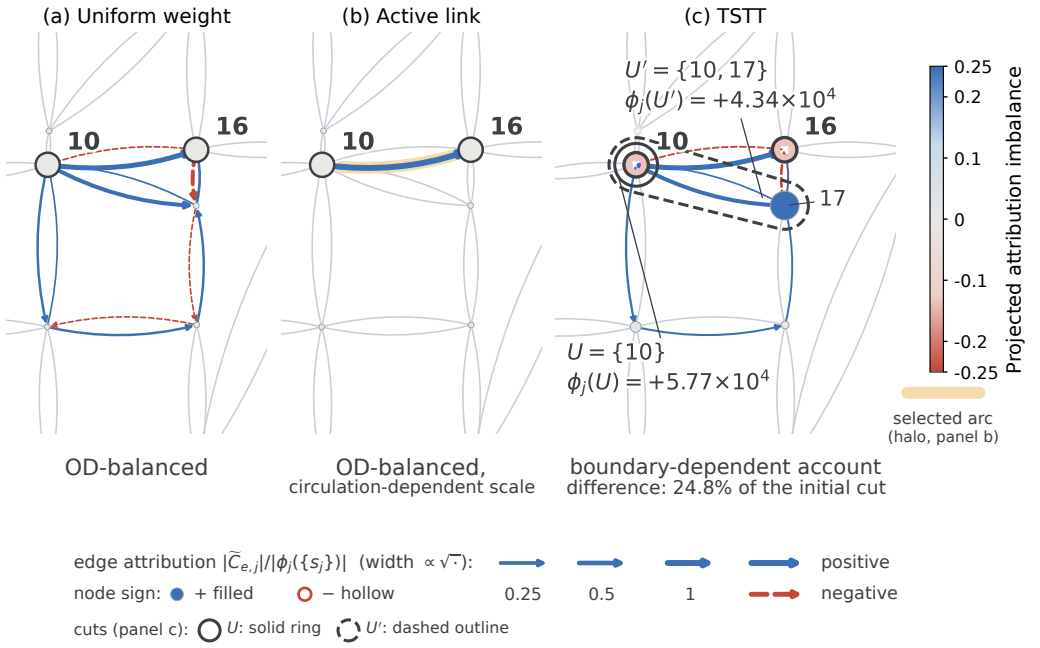


Figure 2: OD $(10, 16)$ endpoint attributions, zero background, learned seed 0. (a) Uniform weights give cut consistency. (b) The active-link outcome balances in the OD direction but has no response-independent scale. (c) The signed TSTT boundary account decreases by 24.8% when the boundary expands from $\{10\}$ to $\{10, 17\}$. Widths encode attribution magnitude; node colors show the projected attribution imbalance $(d_j)_v$ (Eq. (9)) divided by the absolute origin-cut value.

Table 4
TSTT cut ranges as percentages of each account's absolute origin-cut value. A: zero background; B: frozen background flows; C: background demands fixed, all traffic re-equilibrated. Learned values are five-seed medians (ranges); all SUE columns are direct (41 quadrature points, implicit sensitivities).

	Unrestricted range (%)				Connected (%)
OD	Learned A	SUE A	SUE B	SUE C	SUE C
$(17, 10)$	59.71 (58.92–61.23)	59.95	56.38	54.77	35.27
$(10, 17)$	37.55 (37.40–37.94)	37.48	43.62	49.11	31.57
$(16, 10)$	36.80 (36.41–38.40)	36.86	39.99	166.26	127.72
$(10, 16)$	26.85 (26.68–27.07)	26.94	34.02	109.04	83.80
$(15, 10)$	0.84 (0.69–1.16)	0.77	21.98	84.05	58.24
$(10, 15)$	0.81 (0.76–0.88)	0.77	18.89	71.88	49.90
$(10, 11)$	0.23 (0.10–0.35)	0.0075	1.17	84.67	51.27
$(11, 10)$	0.18 (0.03–0.27)	0.0073	0.79	81.09	48.98

For OD $(17, 10)$, the normalized range decreases from 59.95% in A to 54.77% in C while the absolute range increases from 17,607.8 to 123,202.1 TSTT units, about sevenfold: a lower percentage need not mean a smaller absolute difference.

6.3. Separating the background route response

Comparing A, B, and C changes both congestion and sensitivities. To isolate the background response at the same state, split total flow in setting C into the sum of the eight study ODs and the other 520 background ODs. Then $BJ_j^{\text{study}} = \mu_j$ and $BJ_j^{\text{bg}} = 0$. Weighting both sensitivities, at each quadrature point, by the same total-flow TSTT gradient gives

$$\widetilde{C}_j^{\text{tot}} = \widetilde{C}_j^{\text{study}} + \widetilde{C}_j^{\text{bg}}. \tag{18}$$

At node 15 for $(10, 16)$, the background term contributes 49,103.4 against the study term's −973.1 TSTT units to internal node divergence. A vehicle-flow circulation can therefore dominate cost-weighted imbalance (full node breakdown in Appendix C.2, Table C.2). For a concrete comparison, retain the connected regions $U = \{10\}$ and $U' = \{10, 17\}$ from the zero-background illustration — specified local regions, not range-attaining ones — whose complements are also connected and which separate both queries reported next. With $g \in \{\text{study}, \text{bg}, \text{tot}\}$,

$$\Delta\phi_j^{(g)} = -\big(\mathbf{1}_{U'} - \mathbf{1}_U\big)^\top B\widetilde{C}_j^{(g)}, \qquad \Delta\phi_j^{\text{tot}} = \Delta\phi_j^{\text{study}} + \Delta\phi_j^{\text{bg}}.$$

At the same equilibrium states, this enlargement changes the $(10, 16)$ account by −3361.1 (−1305.0 study, −2056.1 background; 1.23% of $\phi_j(U)$) and the $(10, 11)$ account by −1461.4 (−269.9 study, −1191.5 background; 0.72%). Thus large extremal ranges can coexist with small differences at selected sections. Specified boundary differences add across components, whereas the absolute ranges satisfy only $S_j^{\text{tot}} \le S_j^{\text{study}} + S_j^{\text{bg}}$.

6.4. Boundary-dependent signs across OD queries

Six queries away from node 10 were pre-specified by taking the shortest and longest free-flow shortest-path ODs within each demand tercile (demands 200–800 veh/h; path times 2–23; Appendix B.5). Their demands move jointly from 0.2 to 1.6 of published values, with all others fixed; all 528 ODs re-equilibrate under the setting-C protocol.

Table 5 reports the result: all six system contributions C_j are positive, but for the three queries with the longest free-flow shortest-path times the minimum–maximum ranges straddle zero in both the unrestricted and the connected family. OD $(1, 19)$, for instance, has an origin-cut account of 1,771.6 TSTT units while its connected-family accounts range from −13,502 to 24,260: the boundary choice can change the sign of the reported account, not merely its magnitude. Negative accounts occur in 17.8%, 2.5%, and 7.3% of the 14,477, 15,191, and 16,943 enumerated connected boundaries of $(1, 19)$, $(1, 15)$, and $(7, 12)$; no connected boundary of the three sign-consistent queries is negative. These are shares of the enumerated family, not the odds of an analyst drawing such a boundary. For the three crossing queries the system contribution exceeds the origin-cut account by factors of twelve to twenty-eight; for the three sign-consistent queries the two agree within a factor of 1.6, so a boundary account is not a rescaled system contribution even when the two are close. OD $(6, 8)$'s unrestricted range of 99.75% (connected 65.22%) shows that the sign is set by the extrema themselves, not by the size of the normalized range.

Table 5
System contributions and spatial ranges for six prespecified queries, setting C (TSTT units). Connected: all admissible connected regions; screenline: the geometric family constructed independently of attribution values. All C_j are positive. Origin-cut accounts, negative shares, and neighbourhood ranges are retained in Appendix C.5.

OD	C_j	Connected [min, max]	Screenline [min, max]
$(7, 18)$	672.9	[634, 710]	[642, 702]
$(1, 19)$	50,097.7	[−13,502, 24,260]	[−2,767, 22,124]
$(4, 5)$	2,826.0	[2,396, 3,452]	[2,652, 3,176]
$(1, 15)$	77,403.2	[−8,286, 40,540]	[511, 36,621]
$(6, 8)$	73,964.9	[35,307, 66,695]	[45,230, 65,286]
$(7, 12)$	117,482.3	[−17,616, 60,029]	[−2,967, 57,780]

Why one connected boundary attains a negative account is visible in the attribution columns: by Eq. (11), sign crossing in the unrestricted family is decided by the interior positive divergence against the origin-cut account, which it exceeds by factors of about 5.8 to 12. For $(1, 19)$, the connected minimum −13,501.6 is attained by a 13-node connected region containing the origin. The free-flow-time neighbourhood at radius 16 also contains 13 nodes, yet its account is +10,756.0. By Eq. (12), their respective net interior divergences are approximately +15,273.1 and −8,984.4 TSTT units: the account depends on the spatial distribution of weighted demand responses within the region, rather than its node count alone (node sets in Appendix C.5).

At the boundary itself, entering links contribute −21,368.8 and leaving links +7,867.3 to the twenty-six-link signed sum, dominated by large *positive* attributions on *entering* links; Figure C.1 maps the region and each link's signed contribution in the appendix. For the joint six-query demand change, $\phi_j(U^-)$ is OD j's contribution to the change in the signed boundary cost sum H_{U^-} (Eq. (6)); the total change of H_{U^-} sums all six varying ODs' contributions, and $C_j = 50{,}097.7 > 0$. Different regions define different functions H_U, so opposite signs are compatible with integrated gradients. The classification addresses when their summaries can serve as one transferable OD account; otherwise Eq. (12) explains their differences.

The BPR marginal cost $g'_e = t_{0,e} + 0.75t_{0,e}(F_e/c_e)^4$ splits the weights into free-flow and congestion-premium parts. Combining this with the study/background response split at each of the same equilibrium states gives, at the fixed reported U_-,

$$\phi_j(U_-) = \phi_j^{\text{ff,study}} + \phi_j^{\text{ff,bg}} + \phi_j^{\text{cong,study}} + \phi_j^{\text{cong,bg}}, \qquad \phi_j^{a,b} = -\Delta q_j \int_0^1 \mathbf{1}_{U_-}^\top B D_\lambda^a J_j^b(\lambda)\, d\lambda.$$

Table 6 locates the negative account differently across queries: the study congestion term dominates for $(1, 19)$, both congestion terms are negative for $(1, 15)$, and background congestion dominates for $(7, 12)$. The combined free-flow part stays positive at all three boundaries. This is an additive explanation at the same states and fixed boundaries, not a causal effect of removing background traffic or congestion. The interior-divergence split in Appendix C.5 concerns a different sum and cannot be substituted for this boundary decomposition.

Table 6
Four parts of the same boundary account at the reported U_- (setting C, six-query protocol; TSTT units; identical 41-point states, responses, and boundaries as Table 5). Study includes all six query ODs; background includes the other 522. Columns cross free-flow/congestion weights with study/background responses. Entries are rounded independently.

OD	ff, study	ff, background	cong, study	cong, background	Total
$(1, 19)$	−361.3	+759.0	−17,357.9	+3,458.6	−13,501.6
$(1, 15)$	+1,541.8	+333.8	−6,537.4	−3,624.2	−8,286.0
$(7, 12)$	+3,897.0	−1,311.1	+125.5	−20,327.6	−17,616.2

The enumerated extrema establish existence within the admissible family; the prescribed boundaries separate sign from magnitude. All tested origin-centred hop-distance and free-flow-time neighbourhood accounts remain positive, yet within those families the largest account is 8.9, 9.0, and 3.4 times the smallest for $(1, 19)$, $(1, 15)$, and $(7, 12)$ (Appendix C.5): sign stability does not mean that simple, network-distance-defined boundaries give comparable magnitudes. Straight screenlines sharpen the picture. Projecting the public node coordinates onto 36 pre-fixed directions (5° apart, both sides of each line kept, positions midway between adjacent node projections, connectedness and deduplication as in the connected family; no attribution value enters the construction; Appendix C.5) yields 23–152 distinct straight-line boundaries per query. On these, $(1, 19)$ and $(7, 12)$ again attain negative accounts (minima −2,767 and −2,967 TSTT units, on 2 of 152 and 10 of 148 boundaries), shallower than the enumerated minima, while the remaining four queries in the screenline family stay positive with magnitudes spanning up to a factor of 72 for $(1, 15)$ (511–36,621). Practical straight-line sections therefore do not inherit the sign safety of the neighbourhood families, nor do their positive accounts agree in magnitude. Appendix C.5 reports the path-set, dispersion, single-OD, and narrow-band checks. The separable TSTT structure gives a particularly direct interpretation when only one demand changes:

$$\widetilde{C}_{e,j} = g_e(F_e^1) - g_e(F_e^0), \qquad g_e(F_e) = F_e t_e(F_e). \tag{19}$$

Indeed $dF_e/d\lambda = \Delta q_j \partial_j f_e$ along that path, so the fundamental theorem of calculus applies link by link, including background rerouting. Consequently $\phi_j(U) = H_U(f(q^1)) - H_U(f(q^0))$: a negative single-OD boundary account is an actual decrease in that signed boundary cost sum. This identity also checks the numerical edge attribution directly.

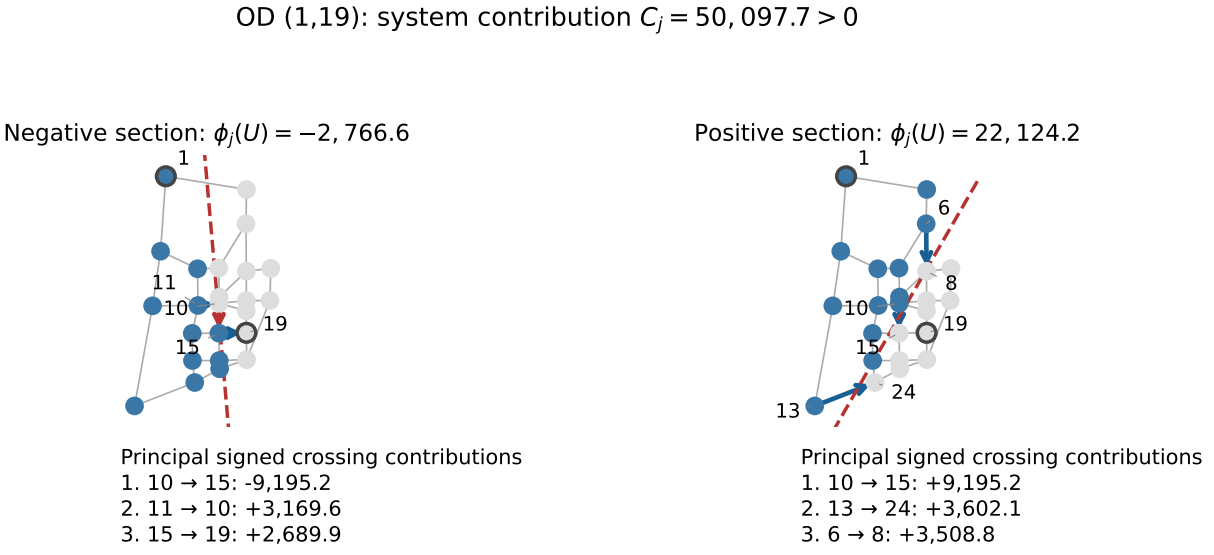


Figure 3: Two straight screenlines for OD $(1, 19)$, setting C. Blue nodes belong to U; grey nodes are outside; dashed lines separate the two sides. Node 1 is inside and node 19 outside in both panels. The two accounts have opposite signs while the network-wide contribution is positive. The lines are the minimum and maximum of the prespecified geometric family; the complete irregular connected minimum and its crossing-link decomposition appear in Figure C.1.

Table 7
Input-stratified coverage across all 18 queries. Spreads use the Sioux Falls cost units. Ratios compare the pre-specified geometric family with all separating regions.

Interval	Median geometric spread	Median unrestricted spread	Median ratio
±10%	3292.5	7359.7	0.574
±20%	6585.0	14719.5	0.574

6.5. Input-stratified OD coverage

An input-only stratified selection retains 18 of 528 positive Sioux Falls OD pairs, with two queries in each demand-by-free-flow-time tertile cell. Each OD varies alone over ±10% and ±20% intervals; other demands remain fixed with endogenous rerouting. The single-OD endpoint identity gives all edge accounts from four endpoint equilibria per query. Appendix C.9 reports the selection rule and all 18 queries. The geometric spread remains below the unrestricted spread for every selected query at both widths. The regional family therefore changes the magnitude of observed spatial variation. This fixed stratified sample is descriptive rather than a probability sample of demand conditions; same-state study/background decomposition is evaluated separately.

6.6. External spatial calculation on Anaheim

The public Anaheim network has 416 nodes, 914 directed links and 1,406 positive OD demands (Transportation Networks for Research Core Team, 2024). We select the six largest demands, vary them jointly from 0.8 to 1.2 times their published values, and hold all other demands fixed while their route choices remain endogenous. Each OD uses ten free-flow shortest paths; candidate routes exclude other zone centroids as intermediate nodes (the first through node is 39). This reference calculation evaluates spatial computation and background mechanisms; no Anaheim surrogate is trained. The reference chain with 21 quadrature states took 1278.7 seconds on the local CPU, including route construction and implicit sensitivities; the six unrestricted extrema took 0.0138 seconds after the columns were available. This separates the saved boundary-enumeration cost from reference attribution cost. Maximum path fixed-point residual was 1.1e-11, flow-balance residual 9.1e-12, and relative quadrature completeness discrepancy 3.2e-05. At the common

Table 8
Anaheim reference attribution (absolute accounts in vehicle-minutes per hour; divide by 60 for vehicle-hours per hour): unrestricted accounts and connected regions formed by removing edges on the OD path in two deterministic breadth-first spanning trees rooted at the origin and destination. These topology-only region rules use no attribution values. Ranges use 41-point integrated columns, with no connected-region enumeration.

OD	System	Unrestricted range	Selected connected range	Regions
(4,2)	34664.1	[-12712.6,20514.0]	[238.9,4932.1]	36
(1,2)	16773.6	[-6032.2,9762.0]	[281.2,2964.7]	26
(2,4)	12143.7	[-4374.6,6733.0]	[99.9,1915.3]	36
(3,2)	14706.7	[-5613.8,9052.9]	[172.9,5777.4]	28
(25,4)	6919.3	[-1805.3,3091.8]	[96.9,1212.2]	22
(25,2)	11445.9	[-4157.3,6904.5]	[126.8,2676.4]	16

Table 9
Anaheim full-path differences between two preselected connected regions (vehicle-minutes per hour). FF denotes free-flow costs, C congestion marginal costs, S the six study demands and B all other demands.

OD	FF/S	FF/B	C/S	C/B	Total
(4,2)	-0.00	-0.10	-1338.21	0.04	-1338.28
(1,2)	21.81	17.46	3.96	-15.69	27.54
(2,4)	0.00	0.00	654.99	0.00	654.99
(3,2)	0.00	-0.00	-1622.92	-0.00	-1622.92
(25,4)	-396.00	4.26	-673.82	1.47	-1064.09
(25,2)	-412.24	0.49	-2134.60	0.13	-2546.22

middle state, the six background-response norms range from 0.115 to 0.487, while their maximum node divergence is below 3.1×10^{-15}. Weighting those same responses by the common total-flow marginal costs gives divergence norms 0.116–0.796. Background rerouting therefore remains a nonzero spatial attribution mechanism on this larger network.

For full-path mechanism analysis, order each query's pre-existing BFS regions lexicographically by its sorted node list and select the first, middle and last regions, denoted U_0, U_1, U_2. This rule uses topology and the OD only, without inspecting attribution. Each fixed comparison is decomposed using the same equilibrium states and total-flow marginal costs into free-flow/study, free-flow/background, congestion/study and congestion/background terms. Table 9 reports $\varphi_j(U_2) - \varphi_j(U_0)$; the additional $U_1 - U_0$ comparisons and exact region memberships are archived with the experiment. These are differences for identical fixed regions, not sums of componentwise ranges. These mechanism columns use 41 equally spaced trapezoidal states. The maximum unrounded decomposition discrepancy is 9.1e-13 per edge. Refining from 21 to 41 states changes the node representation by relative Euclidean errors between 5.8e-06 and 5.2e-05 across queries; the largest selected-account change is 0.206 vehicle-minutes per hour (0.00452% relative). This evaluates spatial accounts directly rather than relying on cancellation in scalar completeness. For $(1, 2)$, the background free-flow term is positive whereas its congestion term is negative, and these contributions largely cancel. For most other selected comparisons the background terms are small relative to the study-response congestion term. The mechanism decomposition identifies both cancellation and relative magnitude under common total-flow cost weights.

7. Application to learned assignment

The spatial framework compares explanations produced by learned assignment models with the reference equilibrium. We retain a final comparison of four training objectives from the same five initial models, each continued for 1,200 updates under a shared validation rule. The 24 independent test paths contain six each with one, two, three and six active OD queries. Statistics average active queries and seeds within each path before summarizing paths; training seeds do not increase the independent path count. Complete

Table 10
Endpoint supervision on 24 independent paths (six each with one, two, three and six active OD queries), five seeds. Flow is the five-seed mean relative error on the original held-out demand states. Other entries first average active queries and seeds within each path, then report medians over paths; the last column gives the 90th percentile. All methods share 1,200 continuation updates and a common validation selection rule.

Method	Flow (%)	Column (%)	System (%)	Cut abs.	Cut (%)	Cut P90 (%)
Flow	0.0386	21.61	2.33	270.16	14.53	34.27
Flow + endpoint edge	0.0518	17.14	3.07	255.83	15.50	28.93
Flow + endpoint edge + cut	0.0513	18.02	3.26	289.68	17.18	26.63
Flow + response	0.0388	14.22	3.31	178.72	12.08	28.17

losses, selection rules, costs and supplementary results appear in Appendix B.8; earlier comparisons are consolidated in Appendix C.10. Ordinary response supervision reduces median attribution-column error from 21.61% to 14.22% and boundary error from 14.53% to 12.08% relative to flow supervision, reductions of 34.2% and 16.9%. These improvements belong to established derivative supervision, not to a new training method proposed here. Adding the endpoint boundary term gives 18.02% column error and 17.18% boundary error; its lower boundary 90th percentile does not establish an overall advantage. Endpoint labels took 64.0 seconds versus 53.1 seconds for center-state solves and response labels. The comparison demonstrates that the spatial criteria distinguish explanation quality beyond flow fit, while providing no overall accuracy or label-cost advantage for the added boundary objective.

8. Conclusions

Spatial OD accounts can be characterized, calculated and interpreted within one framework. Their interchangeability depends jointly on network topology, the query and the admissible region family. For two-sided connected regions, only queried block paths constrain marginal outcome weights, allowing heterogeneity elsewhere. For realized attributions, a node representation supports selected-region comparison, unrestricted extrema, best common-value approximation and exact model disagreement. Reusable projection scans extend this computation to a specified finite family of straight screenlines.

Sioux Falls and Anaheim connect these calculations to traffic adjustment. At common equilibrium states, crossing study and background responses with free-flow and congestion weights identifies additive mechanisms behind boundary differences. Positive network-wide contributions and negative boundary accounts describe different, well-defined outcome changes. Learned assignment provides a further use: spatial criteria reveal explanation errors beyond flow-fit metrics, without requiring a new training algorithm to outperform existing response supervision. The results apply to the fixed road representations, prescribed region families and assignment models stated here, and provide a basis for comparing spatial explanations of traffic demand changes.

Code and data availability

Experiment scripts, the numerical arrays behind the tables and figures, and trained surrogate checkpoints are available from the corresponding author on request. The requested package includes a README mapping results to scripts and arrays. The Sioux Falls and Anaheim networks and demands are available from the public Sioux Falls TNTP repository and Anaheim TNTP repository (Transportation Networks for Research Core Team, 2024); training targets are synthetic fixed-path logit-SUE assignments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A. Mathematical foundations and proofs

A.1. Conventions and conservation transfer

The incidence matrix $B \in \mathbb{R}^{n\times m}$ has entries -1 at an arc's origin and $+1$ at its destination; $\mu_j = e_{t_j} - e_{s_j}$ and $M = [\mu_1, \dots, \mu_K]$. Write $Z = \ker B$ and let $B^\dagger$ be the Moore–Penrose pseudoinverse. All norms below are Euclidean or their induced operator norms unless indicated otherwise. Graphs are loopless multigraphs: parallel and reciprocal arcs remain distinct edges, including in two-edge cycles. Cycle atoms and the block-cut tree are taken on the underlying undirected multigraph. Formally, $\iota(v)$ is the cut-vertex node when v is an articulation vertex and otherwise its unique block node; $\mathcal{P}(s,t)$ lists the atoms on the unique tree path from $\iota(s)$ to $\iota(t)$. A signed simple s–t path vector p satisfies $Bp = \mu$ and the full solution set is $p + Z$. Such a path uses only atoms of $\mathcal{P}(s,t)$ and meets every atom on it.

Signed responses. The circulation and response vectors are signed first-order sensitivities, not nonnegative commodity flows. At a strictly positive flow, sufficiently small signed circulation perturbations also preserve nonnegativity. A restricted response family is treated explicitly below.

Corollary A.1 (IG exchange)**.** *For an exactly conservative C^1 field on an open neighborhood of the attribution path, $BG_{:,j} = \mu_j \Delta q_j$.*

Proof. Differentiate $Bf(q) = Mq$ to obtain $BJ_f(q) = M$, then integrate the jth column along $\gamma(\lambda) = q^0 + \lambda\Delta q$.

A.2. Cycle structure and the classification proofs

Lemma A.2 (structure of circuit classes)**.** *The non-bridge atoms are the non-singleton circuit classes; each bridge is a singleton class. Moreover $Z = \bigoplus_A Z_A$, where $Z_A = \{z \in Z : \operatorname{supp} z \subseteq A\}$, and the summands have disjoint edge supports.*

Proof. No simple cycle crosses an articulation vertex into another block: it would revisit that vertex. Within a non-bridge block, any two edges lie on a common cycle. One way to see this is to subdivide the two edges; the resulting block is still biconnected, so its two new vertices lie on a common cycle, which descends to the required cycle. This is the usual circuit-class characterization of the graphic matroid (Oxley, 2011). Every signed circulation is a linear combination of signed simple-cycle vectors: subtract the fundamental cycles of a spanning forest to leave a circulation on a forest, necessarily zero by successive leaf elimination. Grouping these cycles by their blocks gives the direct sum, and distinct blocks have disjoint edge sets.

Lemma A.3 (circulation-preserving diagonal maps)**.** *For $D = \operatorname{diag}(w)$, $DZ \subseteq Z$ if and only if w is constant on every non-bridge atom; bridge weights are free.*

Proof. For any signed simple-cycle vector z_C, a circulation supported on that cycle has equal signed amplitudes on successive edges, by node balance. Thus $BDz_C = 0$ forces w constant on C, and Lemma A.2 propagates equality throughout each atom. Conversely, atomwise constancy makes D a scalar on every Z_A, so it preserves their direct sum. Bridges carry no circulation.

Proof (Theorem 1). Add a virtual reverse OD edge (t, s) and set

$$\bar{B} = [B, -\mu], \qquad \bar{D} = \operatorname{diag}(w, \alpha).$$

For a fixed α, property (7) is equivalent to $\bar{D} \ker \bar{B} \subseteq \ker \bar{B}$. Indeed, a kernel vector (x, a) obeys $Bx = a\mu$. If $a \neq 0$, apply (7) to x/a to obtain $BDx = a\alpha\mu$. If $a = 0$, subtract the identities for $p + x$ and p, with any

fixed $Bp = \mu$, to obtain $BDx = 0$. These are exactly the two cases of $\bar{B}\bar{D}(x, a) = 0$. Conversely, applying kernel invariance to $(x, 1)$ gives (7) for every $Bx = \mu$, at that same α.

By Lemma A.3, this invariance holds precisely when the augmented weights are constant on its cycle atoms. The virtual edge and a simple s–t path form a cycle meeting every original atom on $\mathcal{P}(s, t)$, so these atoms merge with the virtual edge into one circuit class. No other atoms join it: every new simple cycle contains the virtual edge and a simple s–t path, which uses only those atoms. Other original atoms retain their classes. Thus the augmented criterion is exactly within-atom constancy everywhere and common value α on $\bigcup \mathcal{P}(s, t)$. This proves both directions and identifies $\alpha = w^*$ without changing the universal response family.

Proof (Theorem 2 and Corollary 3). Apply the single-OD theorem to each query at the same w. Necessity requires equality within every atom and among all atoms co-traversed by a queried block path. Their transitive closure is exactly the compatibility-class partition. Conversely, class constancy supplies both single-OD conditions for every query. Along the attribution path, exact conservation gives $BJ_f(\gamma(\lambda))\mathbf{e}_j = \mu_j$; the static theorem therefore yields

$$B\widetilde{C}_{:,j} = \Delta q_j \int_0^1 BD_\lambda J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda = \Delta q_j \left(\int_0^1 w^*_{k(j)}(\lambda) \, d\lambda \right) \mu_j.$$

Corollary A.4 (all-OD class collapse)**.** *Requiring the criterion at one common w for every ordered pair of distinct vertices makes all atoms in a connected component share one constant.*

Proof. For two atoms incident to the same cut vertex, choose endpoints on their opposite sides. Their OD block path contains both atoms, forcing their constants to agree. Connectivity of the block-cut tree propagates the equality throughout the component.

Remark A.1 (a common flow realization). The merging proof requires one common weight vector. Computing the queries at different realized flows does not in general supply this premise.

A.3. Direction, integration, and network representation

Proposition A.5 (universal directional criterion)**.** *Let $\hat{x}_j = B^\dagger \mu_j$, let N span Z with orthonormal columns, and let P_{μ_j} project orthogonally onto* $\mathrm{span}\{\mu_j\}$*. At fixed w, universal directional balance is equivalent to*

$$(I - P_{\mu_j})BD\hat{x}_j = 0, \qquad (I - P_{\mu_j})BDN = 0.$$

The scale may depend on the response. Requiring one common boundary value additionally requires $BDN = 0$. If directional balance holds along the whole path for a conservative model, its integrated column balances.

Proof. Every response is $\hat{x}_j + z$, $z \in Z$. Apply $I - P_{\mu_j}$ to $BD(\hat{x}_j + z)$ and vary z. The scale is $\mu_j^\top BD(\hat{x}_j + z)/\|\mu_j\|^2$ and is independent of z precisely when $BDZ = 0$. Integrating the realized directional identity gives the endpoint conclusion. Equivalently, invariance on all incidence fibers, $Bx = By \Rightarrow BDx = BDy$, is exactly $DZ \subseteq Z$, since $x - y$ ranges over Z.

Example A.6 (strict separation of the criteria)**.** *On the diamond $(s, a), (a, t), (s, b), (b, t)$ with weights $(1, 1, 1, 2)$, the upper-path response $(1, 1, 0, 0)$ has divergence μ after weighting, but the lower-path response has weighted divergence $(-1, 0, -1, 2)$ in node order (s, a, b, t), outside* $\mathrm{span}\{\mu\}$*. On the triangle $(s, t), (t, u), (u, s)$ with weights $(1, 0, 0)$, every response $x = (1 + c, c, c)$ satisfies $BDx = (1 + c)\mu$; direction balances, but the common boundary value fails because the circulation $(1, 1, 1)$ maps to $\mu \neq 0$.*

If directional balance holds for all queries at common w, differences give $BDZ \subseteq \bigcap_j \mathrm{span}\{\mu_j\}$. For non-collinear queries this intersection is $\{0\}$, so directional balance implies the common-value property. This uses the full response space Z.

Restricted families. For $S \subseteq Z$ with basis N_S, test responses in $x_j^{\mathrm{ref}} + S$, where $Bx_j^{\mathrm{ref}} = \mu_j$. The same proof gives the direction tests with $\hat{x}_j, N$ replaced by x_j^{ref}, N_S; common boundary value replaces the second test by $BDN_S = 0$. Using $\hat{x}_j$ requires $\hat{x}_j - x_j^{\mathrm{ref}} \in S$. For a fixed-path model $f = \Delta h$, $A_{\mathrm{OD}}h(q) = q$ and $B\Delta = MA_{\mathrm{OD}}$. Choose a path vector a_j with $A_{\mathrm{OD}}a_j = \mathbf{e}_j$ and set $x_j^{\mathrm{ref}} = \Delta a_j$. Differentiation yields

$$J_f \mathbf{e}_j - x_j^{\mathrm{ref}} = \Delta(\partial_j h - a_j) \in \Delta \ker A_{\mathrm{OD}} =: T_{\mathrm{swap}}.$$

This space includes same-OD path exchanges of every OD: background ODs may reroute even when their demands remain fixed.

Example A.7 (endpoint cancellation)**.** *For two series bridges, $f(q) = (q, q)$ and $T(F) = F_1 + F_2 + [\sin(2\pi F_1) - \sin(2\pi F_2)]/(2\pi)$ give*

$$\widetilde{C}_1 - \widetilde{C}_2 = \frac{\sin(2\pi q^1) - \sin(2\pi q^0)}{\pi}.$$

For $q^0 = 0$, the nonzero balanced endpoints in $[0, 1]$ are $q^1 = 1/2, 1$, although the two marginal weights differ almost everywhere. At $q^1 = 0.3$ balance fails. Endpoint cancellation therefore does not establish the pointwise classification conditions.

Example A.8 (network subdivision)**.** *A single edge with $f(q) = q$, $T(F) = \tau F$, $\tau > 0$, has cut account $\tau\Delta q$. Subdivide it into $(s, v), (v, t)$ and set $f'(q) = (q, q)$, $T'(F_1, F_2) = \tau[\alpha F_1 + (1-\alpha)F_2]$, $0 < \alpha < 1$. The composite demand-to-outcome map is unchanged, but $\widetilde{C}' = \tau\Delta q(\alpha, 1-\alpha)$ and $(B\widetilde{C}')_v = \tau(2\alpha - 1)\Delta q$. The two cuts account $\tau\alpha\Delta q$ and $\tau(1-\alpha)\Delta q$. They agree only for $\alpha = 1/2$, when each is half the undivided account; their sum remains the scalar contribution $\tau\Delta q$. The invariance studied here thus concerns conservative responses on a fixed network with a fixed edgewise outcome definition, rather than invariance under subdivision.*

A.4. Approximate conservation and attribution balance

Theorem A.9 (Failure of attribution balance under approximate conservation)**.** *Fix $\varepsilon > 0$ and $\delta \neq 0$. With input dimension $K = 2$ and $v \in \mathrm{im}(B)$, $\|v\| = 1$, there is an analytic demand-to-flow map $f_\varepsilon : \mathbb{R}^2 \to \mathbb{R}^m$ (constructed in Appendix A) such that*

$$\sup_{q \in \mathbb{R}^2} \|r_\varepsilon(q)\| \leq \varepsilon, \qquad \text{with } r_\varepsilon \equiv 0 \text{ on the diagonal set } \{q : q_1 = q_2\},$$

yet along the diagonal path $q(\lambda) = \lambda(\delta, \delta)$ its per-column attribution residuals are

$$R_1 = v\,\delta, \qquad R_2 = -v\,\delta : \text{ fixed scale } |\delta|, \text{ independent of } \varepsilon.$$

Proof of Theorem A.9. For $K = 2$ and $\|v\| = 1$, $v \in \mathrm{im}\, B$, set

$$r_\varepsilon(q) = \varepsilon v \tanh((q_1 - q_2)/\varepsilon), \qquad f_\varepsilon(q) = B^\dagger(Mq + r_\varepsilon(q)).$$

Since $BB^\dagger$ projects onto $\mathrm{im}\, B$, the residual of this analytic field is exactly r_ε, bounded globally by ε and zero on $q_1 = q_2$. Writing $x = (q_1 - q_2)/\varepsilon$ gives $J_{r_\varepsilon} = v\,\mathrm{sech}^2(x)(1, -1)$. On $q(\lambda) = \lambda(\delta, \delta)$, $x = 0$, so the residual identity (17) yields $R_1 = \delta v$, $R_2 = -\delta v$. The second derivatives are bounded above by C/ε and attain magnitude c/ε at a fixed nonzero x; thus $\mathrm{Lip}(J_{r_\varepsilon}) \asymp \varepsilon^{-1}$. □

Proposition A.10 (strictly positive strengthening)**.** *Let $K \geq 2$, $Q = [\ell, u]^K \subset (0, \infty)^K$, and let f_0 be exactly conservative and C^1 near Q, with $\varrho = \min_{q \in Q, e}(f_0(q))_e > 0$. For unit $v \in \mathrm{im}\, B$, define $f_\varepsilon = f_0 + B^\dagger r_\varepsilon$ with the same r_ε as above. Its residual is bounded by ε and vanishes on $q_1 = q_2$. If $\varepsilon < \varrho/\|B^\dagger\|_2$, all flows stay strictly positive. Along any path $q^0 + \lambda\delta(\mathbf{e}_1 + \mathbf{e}_2) \subset Q$ with $q_1^0 = q_2^0$, its column residuals are $R_1 = \delta v$, $R_2 = -\delta v$, independent of ε.*

Proof. The projection identity gives the residual exactly, while $\|f_\varepsilon - f_0\|_\infty \leq \|B^\dagger\|_2 \varepsilon < \varrho$ preserves positivity. On the diagonal path the first two partial derivatives of the residual are $v, -v$. Integrate them; the exactly conservative f_0 contributes zero residual. The normalized residual is therefore $\|R_j\|/(|\Delta q_j| \|\mu_j\|) = 1/\|\mu_j\|$.

Theorem A.11 (local neighborhood bound)**.** *Let $r \in C^{1,1}(\mathcal{N};\mathbb{R}^n)$ on an open neighborhood of the attribution path, with $\mathrm{Lip}(J_r) \le H$. Suppose, for some $\rho > 0$ and every j, λ, $\gamma(\lambda) + [-\rho,\rho]\mathbf{e}_j \subset \mathcal{N}$ and $\|r(\gamma(\lambda) \pm \rho\mathbf{e}_j)\| \le \varepsilon$. Then*

$$\|R_j\| \le |\Delta q_j|(\varepsilon/\rho + H\rho/2).$$

Proof of Theorem A.11. For $q = \gamma(\lambda)$, the hypotheses place $q + [-\rho,\rho]\mathbf{e}_j$ inside $\mathcal{N}$. Taylor's formula with a Lipschitz Jacobian gives $r(q \pm \rho\mathbf{e}_j) = r(q) \pm \rho\partial_j r(q) + \xi_\pm$, $\|\xi_\pm\| \le H\rho^2/2$. Subtracting the two formulas and using the endpoint residual bounds yields $\|\partial_j r(q)\| \le \varepsilon/\rho + H\rho/2$. Integrating (17) proves the claim. □

Remark A.2 (Sharpness of the square-root rate). The rate in Theorem A.11 cannot be uniformly improved within its own hypothesis class. With $K = 2$, unit $v \in \mathrm{im}(B)$, and

$$r_\varepsilon(q) = \varepsilon v \sin\big((q_1 - q_2)/\sqrt{\varepsilon}\big), \qquad f_\varepsilon(q) = B^\dagger\big(Mq + r_\varepsilon(q)\big),$$

the projection identity of the proof of Theorem A.9 gives output residual exactly r_ε, so $\sup_q \|r_\varepsilon(q)\| \le \varepsilon$. Moreover

$$J_{r_\varepsilon}(q) = \sqrt{\varepsilon}\, v \cos\big((q_1 - q_2)/\sqrt{\varepsilon}\big)(1,-1), \qquad \mathrm{Lip}(J_{r_\varepsilon}) \le 2$$

for every ε: the hypotheses hold with $\mathcal{N} = \mathbb{R}^2$, $H = 2$, and any $\rho > 0$, uniformly in ε. Along the diagonal path $q(\lambda) = \lambda(\delta,\delta)$ the derivative evaluates to $\sqrt{\varepsilon}\, v(1,-1)$, so the residual identity (17) gives

$$R_1 = \delta\sqrt{\varepsilon}\, v, \qquad R_2 = -\delta\sqrt{\varepsilon}\, v,$$

of exact order $\sqrt{\varepsilon}$. Within the class of residuals with residual bound ε and bounded Jacobian Lipschitz constant, no uniform $o(\sqrt{\varepsilon})$ refinement of Theorem A.11 is therefore possible; the exponent, not merely the constant, is sharp.

Lemma A.12 (minimum-norm support)**.** *For a signed s–t path p, $\hat{x} = B^\dagger\mu = (I - P_Z)p$ is supported on $\bigcup \mathcal{P}(s,t)$.*

Proof. The minimum-norm point of $p + Z$ is its orthogonal projection onto $Z^\perp$. Lemma A.2 gives $P_Z = \sum_A P_{Z_A}$ with disjoint supports; both p and every $P_{Z_A}p$ vanish on off-path atoms.

Proposition A.13 (combined dispersion–residual bound)**.** *Let $f \in C^1$ near the attribution path, $r = Bf - Mq - b$ with the constant background balance b of Section 5.2 ($b = 0$ without background), and let $w(\lambda) = \nabla T(f(\gamma(\lambda)))$ be continuous. Let $\bar{w}(\lambda)$ be the compatibility-class mean vector and $\eta(\lambda) = \|w(\lambda) - \bar{w}(\lambda)\|_\infty$. Writing $J_j(\lambda) = J_f(\gamma(\lambda))\mathbf{e}_j$ and $d_j = (I - P_{\mu_j})B\widetilde{C}_{:,j}$, we have*

$$\|d_j\|_2 \le |\Delta q_j|\|B\|_2 \int_0^1 \left[\eta(\lambda)\|J_j(\lambda)\|_2 + \|\bar{w}(\lambda)\|_\infty \|B^\dagger\|_2 \|\partial_j r(\gamma(\lambda))\|_2\right] d\lambda.$$

Proof. Differentiation gives $BJ_j = \mu_j + \partial_j r$ — b is constant and drops out of the derivative — so $J_j = \hat{x}_j + z_j + B^\dagger\partial_j r$, $z_j \in Z$. Let $\bar{D} = \mathrm{diag}(\bar{w})$. Class constancy gives $B\bar{D}z_j = 0$ by Lemma A.3. By Lemma A.12, $\hat{x}_j$ is supported on the OD block path, whose atoms lie in one compatibility class $k(j)$; hence $B\bar{D}\hat{x}_j = \bar{w}_{k(j)}\mu_j$. Therefore

$$(I - P_{\mu_j})BDJ_j = (I - P_{\mu_j})B\left[\mathrm{diag}(w - \bar{w})J_j + \bar{D}B^\dagger\partial_j r\right].$$

The projector is non-expansive, the two diagonal operator norms are η and $\|\bar{w}\|_\infty$, and integration gives the bound. For exact conservation only the dispersion term remains; for $\eta = 0$ only the derivative residual remains. Theorem A.11 can bound that derivative term under its local hypotheses.

A.5. Exact boundary error and its path bound

Proof. Set $b = Be_j$ and $I = V \setminus \{s_j, t_j\}$. Since $\mathbf{1}^\top b = 0$, Eq. (11) applied to e_j gives an interval of extrema with midpoint

$$-b_{s_j} - \frac{1}{2}\sum_{v\in I} b_v = \frac{1}{2}(b_{t_j} - b_{s_j})$$

and half-width $\frac{1}{2}\sum_{v\in I}|b_v|$. The larger absolute endpoint is the absolute midpoint plus this half-width, proving Eq. (15). This argument concerns the attained extrema; it does not require every value between them to be attained. The map $\mathcal{K}_j$ is a seminorm because it is a sum of absolute linear functionals. Moreover $e_j = \Delta q_j \int_0^1 h_j(\gamma(\lambda))\, d\lambda$. Its homogeneity and triangle inequality therefore yield

$$\epsilon_j^{\mathcal{U}} \le \mathcal{K}_j(e_j) \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda,$$

proving the path statement. Its kernel is $\ker B$: if the seminorm vanishes, all interior divergences vanish and $b_{s_j} = b_{t_j}$; the zero sum then makes both endpoints zero.

A.6. Connected-region classification

We retain the loopless multigraph conventions of Appendix A.1; reciprocal arcs are distinct undirected edges when defining atoms and connectivity.

Lemma A.14 (localization of a separating bond)**.** *If $G[U]$ and $G[V \setminus U]$ are connected, all edges of $\delta(U)$ lie in one cycle atom (or its singleton bridge). If $s \in U$ and $t \notin U$, that atom lies on $\mathcal{P}(s,t)$.*

Proof. For two distinct crossing edges, connect their endpoints by a simple path inside U and by another inside its complement. The two paths have disjoint vertex sets; together with the crossing edges they form a simple cycle, including the possible two-edge cycle. Hence all crossing edges share a circuit class. A singleton crossing edge is a bridge. A simple s–t path crosses this boundary, so it meets its atom; every atom met by such a path lies on $\mathcal{P}(s,t)$.

Lemma A.15 (connected prefixes in an atom)**.** *Let A be a non-bridge atom and let $a \neq b$ be two of its vertices. Its vertices admit an ordering $v_1 = a, \ldots, v_k = b$ such that every proper nonempty prefix and its suffix induce connected subgraphs. The assertion includes two-vertex parallel-edge atoms.*

Proof. For two vertices the assertion is immediate. Otherwise add an a–b edge temporarily if needed. The connected-prefix property is the classical st-numbering property (Even and Tarjan, 1976); we give a construction here. A biconnected graph with this edge admits an open-ear construction starting from a cycle containing that edge. For completeness, this construction follows directly from vertex connectivity. If vertices remain outside the current subgraph, any component of those vertices has at least two distinct attachment vertices in the current subgraph: a unique attachment would be an articulation of the original graph. A path through the component between two attachments supplies the next ear. Repeat until all vertices have entered. Edges left over are irrelevant to the ordering. The initial cycle exists because the added edge is not a bridge.

Order the other arc of the initial cycle from a to b. For each subsequent ear, let u precede v among its two attachment vertices; insert its internal vertices, in their order from u to v, immediately after u. Every new internal vertex has an earlier and a later neighbor. Existing vertices keep these neighbors. The first and last vertices remain a, b. Every vertex except a therefore has an earlier neighbor and every vertex except b a later neighbor. Repeated steps along these neighbors connect every prefix to a and every suffix to b. Removing the temporary edge does not affect this argument: it is never internal to a proper prefix or suffix.

Proof (Theorem 4). For sufficiency, Lemma A.14 places every crossing edge in one traversed atom. Its weight is α, so $-\mathbf{1}_U^\top BDx = -\alpha \mathbf{1}_U^\top Bx = -\alpha \mathbf{1}_U^\top (e_t - e_s) = \alpha$. No other edge enters this boundary sum.

For necessity, fix a traversed atom A. Delete its edges but keep its vertices. The remaining graph has one connected component T_v for each $v \in V(A)$, containing that vertex and no other vertex of A. Indeed

an outside path between two vertices of A would join their blocks in a cycle, contrary to the block-cut tree. Connectivity of G shows these components partition V. There are distinct vertices a, b with $s \in T_a$ and $t \in T_b$; these are the entry and exit of the OD block path in A. For a bridge, take the cut consisting of its s-side component. Since its unit conservative crossing response is fixed, the assumed account immediately gives that bridge's weight α.

For a non-bridge atom, choose the ordering of Lemma A.15. Set $U_i = \bigcup_{\ell \le i} T_{v_\ell}$ for $1 \le i < k$. The bags and the prefix/suffix connectivity show $U_i \in \mathcal{U}_{st}^{\mathrm{conn}}$. Its crossing edges are exactly those of the corresponding prefix in A. Now choose any signed local response y with $B_A y = e_b - e_a$, where B_A is the incidence matrix on A. Extend it by a signed unit path from s to a inside T_a and from b to t inside T_b, with zero on other edges. The result satisfies $Bx = \mu$. Thus the assumption gives

$$-\sum_{\ell \le i} (B_A D_A y)_{v_\ell} = \alpha \quad (1 \le i < k).$$

The first prefix gives $(B_A D_A y)_a = -\alpha$, differences of consecutive prefixes give zero at every interior vertex, and zero total incidence sum gives $(B_A D_A y)_b = \alpha$. Consequently $B_A D_A y = \alpha(e_b - e_a)$ for every local conservative y. The single-OD classification (Theorem 1), applied to the one-atom graph A, forces every edge in A to have weight α. This holds for every traversed atom. The sufficiency argument already shows that no off-path weight is constrained.

Proof (Corollary 5). Apply Theorem 4 at the common w to each query. Each queried path requires one weight on all its edges; intersecting paths propagate equality transitively. Conversely these transitive equalities satisfy every single-query condition. Atoms not on any queried path enter none of these equalities.

A.7. Finite-change supervision and joint-path implication

Proof (Proposition B.1). Write $r \in \{L, S\}$ and $c^r(q) = g(f^r(q))$ componentwise. Along a single-coordinate segment $q(u) = q^- + u(q^+ - q^-)$, the chain rule gives

$$\frac{d}{du} c^r(q(u)) = (q_j^+ - q_j^-) D^r(q(u)) J_j^r(q(u)).$$

Integration gives the attribution column exactly as $\widetilde{C}^r_{:,j} = c^r(q^+) - c^r(q^-) = z_j^r$. Corollary 8 applied to their difference proves the stated maximum-error identity. No reference input derivative is needed to obtain this label.

For the local-to-joint implication set $A(q) = B[c^L(q) - c^S(q)]$. Equality of every separating boundary account of a sufficiently small single-coordinate change implies $\mathcal{K}_j(c^L(q+\epsilon e_j) - c^L(q) - c^S(q+\epsilon e_j) + c^S(q)) = 0$. The seminorm's kernel is $\ker B$: its expression forces all internal entries of Bh to vanish and its two endpoint entries to agree; their zero total then forces both to vanish. Hence $A(q + \epsilon e_j) = A(q)$. Differentiability and division by ϵ yield

$$\partial_j A(q) = B[D^L(q) J_j^L(q) - D^S(q) J_j^S(q)] = 0.$$

On any valid joint attribution path, integration of this identity gives zero divergence of the corresponding attribution-column error, and therefore matching accounts on every separating boundary. This needs the local matching premise throughout the domain containing the path; a finite training sample supplies no such premise.

B. Experimental methods

B.1. Network, path sets, and demand settings

The public Sioux Falls network has 24 nodes and 76 directed arcs (Transportation Networks for Research Core Team, 2024). All arcs belong to one cycle atom, hence one compatibility class. The eight queries are the positive-demand OD pairs with the largest published demands, with ties broken by ascending origin then destination node ID. All happen to involve node 10. The queries are $(10, 16)$, $(16, 10)$, $(10, 11)$, $(10, 15)$, $(15, 10)$, $(10, 17)$, $(11, 10)$, $(17, 10)$, with published demands $q_{\mathrm{base}} = (4400, 4400, 4000, 4000, 4000, 3900, 3900, 3900)$

veh/h. The attribution path changes these demands from $0.2q_{\text{base}}$ to $1.6q_{\text{base}}$. Node IDs in the traffic experiments are one-based. Flows retain the benchmark's approximate-hourly convention; one time unit is 0.01 h. A reported TSTT number is $\sum_e F_e t_e(F_e)$ with F_e in veh/h and t_e in these time units: multiplying by 0.01 gives the hourly vehicle-hour total, without extrapolating to a day. All absolute attribution and boundary-account numbers use the same convention.

For each OD we retain every loopless directed path if enumeration finishes within 200 paths, otherwise the ten shortest by free-flow cost (Yen's algorithm, lexicographic edge-sequence tie breaking); no OD in this network has a fully enumerated set under the threshold. All eight query ODs use ten paths. The link–path incidence Δ and the matrix A summing paths within each OD stay fixed, so $B\Delta = MA$ with the appropriate full or study OD incidence matrix. The full-background ten-path set has 5,280 paths (ten per OD; the expanded set 10,560, twenty per OD); Appendix C.4 compares these candidate-path models.

The path-exchange response space $T_{\text{swap}} = \Delta \ker A$ (Appendix A.3) has dimension 32 in A with ten paths for each study OD, and dimension 53 in C with either ten or twenty paths for every OD, equal to the full circulation-space dimension $m - n + 1 = 53$ (ranks computed from within-OD differences of link–path incidence columns). Full background therefore spans the complete circulation space through allowable path exchanges, while the eight-OD path model spans a proper subspace.

The three demand settings share the same query endpoints. In setting A, all other demands are zero. In setting B, assign all positive background demands jointly with the study demands fixed at $q^0 = 0.2q_{\text{base}} = (880, 880, 800, 800, 800, 780, 780, 780)$ veh/h in the listed query order, freeze the resulting background link flows, and re-equilibrate only the eight query ODs. In setting C, the other 520 positive OD demands stay at published values while all 528 ODs re-equilibrate jointly at every path point and perturbation. TSTT and its gradient use total link flows in B and C; setting C is a full-background assignment with eight attribution queries, not an attribution evaluation of all 528 ODs.

B.2. Equilibrium and its sensitivities

Unless stated otherwise, the fixed-path logit stochastic user equilibrium (SUE) uses $\theta = 1$ and BPR costs $t_e(F_e) = t_{0,e}[1 + 0.15(F_e/c_e)^4]$. With Π_j the path set,

$$h_p = q_j p_p(F), \quad p_p(F) = \frac{\exp[-\theta(\Delta^\top t(F))_p]}{\sum_{r\in\Pi_j} \exp[-\theta(\Delta^\top t(F))_r]}, \quad F = \Delta h. \tag{B.1}$$

This is the solution of

$$\min_{h\geq 0,\ Ah=q} \left\{ \sum_e \int_0^{(\Delta h)_e} t_e(s)ds + \theta^{-1} \sum_p h_p(\log h_p - 1) \right\}.$$

For positive demands the feasible set is nonempty and compact and the objective strictly convex. A zero path flow can be improved by transferring mass from a positive path of the same OD, since the entropy change is $\theta^{-1}\epsilon \log \epsilon + O(\epsilon) < 0$; hence the minimizer is positive. Its Hessian and sensitivity equations are

$$H = \Delta^\top \operatorname{diag}(t_e')\Delta + \theta^{-1} \operatorname{diag}(1/h_p) \succ 0, \qquad \begin{bmatrix} H & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \partial_j h \\ \partial_j \nu \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{e}_j \end{bmatrix}. \tag{B.2}$$

A has full row rank, so the system is nonsingular and the implicit function theorem gives a C^1 demand-to-flow map.

For background re-routing we solve the equivalent total-link-flow fixed point $F = L(F, q) = \Delta\{q_j p_{\Pi_j}(F)\}_j$ by Newton steps with residual line search. For frozen background the corresponding equation is $F = F^0_{\text{bg}} + \Delta_{\text{study}}\{q_j p_{\Pi_j}(F)\}_{j\in\text{study}}$, where F^0_{bg} is fixed at the joint q^0 equilibrium. Costs and gradients use this total F in both equations. Every solution is checked against its path-loading equation, with maximum absolute path-flow gap below 10^{-7} veh/h. The 76-dimensional implicit system avoids forming a dense path Hessian. For an OD group R, define

$$S_R = \theta \sum_{k\in R} \Delta_k \left[\operatorname{diag}(h_k) - h_k h_k^\top / q_k\right] \Delta_k^\top, \quad W = \operatorname{diag}(t_e'(F_e)), \quad S = S_{\text{all}}.$$

Differentiating the logit probabilities gives $\partial_F L_R = -S_R W$. Consequently the total sensitivity satisfies

$$(I + SW)J_j = \Delta_j p_j, \qquad J_j^{\text{study}} = \Delta_j p_j - S_{\text{study}} W J_j, \qquad J_j^{\text{bg}} = -S_{\text{bg}} W J_j. \tag{B.3}$$

Here "study" includes all eight queried ODs and "background" the other 520: $J_j = J_j^{\text{study}} + J_j^{\text{bg}}$, $BJ_j^{\text{study}} = \mu_j$, $BJ_j^{\text{bg}} = 0$, the study term including responses of the other seven study ODs. The decomposition uses one joint equilibrium, rather than sensitivities at different congestion states. For setting B the background flows have no derivative, so the same implicit system uses $S = S_{\text{study}}$, with $J_j^{\text{study}} = J_j$ and $J_j^{\text{bg}} = 0$.

Central finite differences independently re-solve Eq. (B.1) at $q \pm a_j \mathbf{e}_j$, $a_j = 0.005 q_{\text{base},j}$ and $a_j/2$; both steps are checked at every $\theta = 1$ ten-path re-routing integration point, the path-set and θ variations add half-step checks for all eight columns at the path midpoint, and the link Newton solver is compared against the successive-averages background equilibria at their common eleven quadrature points.

B.3. Conservative learned surrogate

The zero-background surrogate parameterizes $h_j = q_j \,\text{softmax}(z(q)|_{\Pi_j})$, $f = \Delta h$. The three hidden layers each have 128 tanh units; the final layer is linear. The logit head uses training-standardized demands while the raw q_j multiplies the shares, preserving nonnegative flow and exact conservation. Computations use double precision. Training targets are fixed-path logit-SUE assignments ($\theta = 1$, path fixed-point tolerance 10^{-6}). The 2500/857/857 training/validation/test split uses seed 20260820, with demands $q_j = q_{\text{base},j} s_j$ and independent s_j drawn from $\text{LogNormal}(0, 0.5^2)$ clipped to $[0.2, 2]$. Adam uses learning rate 0.003, full-batch link-flow MSE, at most 12,000 steps, and validation every 50 steps. The minimum-validation-error state of each seed $0, \ldots, 4$ supplies the reported results, with automatic differentiation supplying their Jacobians; the test split is used only for final reporting.

B.4. Integration grids, connected boundaries, and cancellation diagnostics

All numerical integrations use composite trapezoids. The principal learned zero-background results use 201 quadrature points; direct zero-background SUE and both background settings use 41 in the main table. Frozen background compares 11 with 41 points; re-routing compares 11, 21, and 41 nested points; the two near-balanced queries add 81 and 161 nested points; the learned-versus-SUE sensitivity comparison uses identical 41-point grids, kept separate from the 201-point learned results (Appendix C.1).

Connected boundaries require both $G[U]$ and $G[V \setminus U]$ to be connected on the underlying undirected graph, with $s_j \in U$, $t_j \notin U$. Exact enumeration yields 10,270–13,253 admissible sets for each of the eight main study ODs, and 4,621–16,943 for the six additional queries of Appendix B.5. Extrema of $-\mathbf{1}_U^\top B \widetilde{C}_j$ are evaluated by exact enumeration, cross-checked against independent subset enumeration on restricted neighborhoods; attaining sets are verified for membership with values recomputed by direct summation. These are connected regions, not claimed to be administrative boundaries.

For cancellation diagnostics define $A_j = |\Delta q_j| \int_0^1 \sum_{v \notin \{s_j, t_j\}} |(BD_\lambda J_j)_v| d\lambda$ and $\kappa_j = 1 - S_j/A_j$ when $A_j > 0$. The triangle inequality gives $0 \le S_j \le A_j$. Both quantities use the specified 201-point learned grid and are summarized over the five seeds; checks on additional path-set or dispersion models sample the demand-path midpoint and endpoints.

B.5. Six pre-specified queries away from node 10

The selection rule uses only published demands and free-flow shortest-path times. Candidates are the positive-demand ODs whose origin and destination both differ from node 10; ranking them by demand and splitting into three equal-count terciles, the minimum- and maximum-free-flow-shortest-path OD within each tercile is selected, with ties broken by ascending (origin, destination). The six queries are $(7, 18)$, $(1, 19)$, $(4, 5)$, $(1, 15)$, $(6, 8)$, and $(7, 12)$, with published demands $(200, 300, 500, 500, 800, 700)$ veh/h and free-flow shortest-path times $(2, 22, 2, 23, 2, 19)$ time units; the joint protocol, sensitivities, and boundary enumeration are as in Section 6.4, with conservation checks at setting-C floating-point levels (maximum path fixed-point residual 1.6×10^{-10} veh/h, midpoint implicit-versus-finite-difference column error 2.6×10^{-8}, completeness 4.3×10^{-7}).

For the single-OD checks, let $\bar{q} = q^{\text{base}}$ be the full published demand vector over all 528 positive-demand ODs. For each focal query $j \in \{(1,19),(1,15),(7,12),(6,8)\}$, define

$$q^{(a)} = \bar{q} + (aq_j^{\text{base}} - \bar{q}_j)e_j, \qquad q^{(b)} = \bar{q} + (bq_j^{\text{base}} - \bar{q}_j)e_j, \qquad (a,b) \in \{(0.2,1.6),(0.9,1.1),(0.8,1.2)\}, \quad \text{(B.4)}$$

where e_j is the coordinate unit vector. All other 527 demands, including the other five selected queries, remain at their published values $\bar{q}_{-j}$ along the segment. Direct equilibria and the learned small-band checks use these same reference states. The setting-C solver, path sets, and boundary definitions are unchanged; checks and interval results are in Appendix C.5. For each interval, Eq. (19) gives the attribution column from the two endpoint equilibria alone.

B.6. Anaheim network, equilibrium and predefined regions

Both network and demand files are the public Transportation Networks for Research Core Team Anaheim TNTP data (Transportation Networks for Research Core Team, 2024). The network header supplies free-flow times in minutes and capacities in vehicles per hour; all roads use the published BPR coefficient 0.15 and power 4. We use $t_e(F_e) = t_e^0[1 + 0.15(F_e/c_e)^4]$ and TSTT marginal weight $t_e^0[1 + 0.75(F_e/c_e)^4]$. The six study OD demands, ranked by descending base demand and then lexicographic OD, are $(4,2)$: 2106.7, $(1,2)$: 1365.9, $(2,4)$: 1271.4, $(3,2)$: 1237.9, $(25,4)$: 1232.4, $(25,2)$: 1232.3 vehicles per hour. They move jointly from 0.8 to 1.2 times base demand; all remaining positive demands stay at their published base values and retain endogenous route choice. Every OD uses ten loopless free-flow shortest routes from Yen's algorithm. Ties within 10^{-12} minutes are resolved by lexicographic edge-index sequence in the TNTP road order. Nodes 1–38 are centroids and cannot be intermediate route nodes; the first through node is 39. Logit scale $\theta = 1$ minute^{-1} applies to route travel time in minutes. The edge Newton solver targets maximum loading residual below 10^{-8} vehicles per hour, uses at most 80 iterations and backtracking up to 30 halvings per step, and checks the original path loading residual below 10^{-7}. Implicit derivatives solve the same fixed-point equation. Trapezoidal integration uses equally spaced states, with warm starts from a neighboring saved state. The 41-state refinement reuses all 21 old states and computes only the twenty new midpoints. For each study OD, construct undirected BFS spanning trees rooted at each endpoint. At every visited node, traverse neighbors in increasing node-index order. Deleting each edge on the tree's endpoint path creates two connected parts; orient the region to contain the OD origin, and remove duplicate regions. This gives the selected connected family, not all connected separating regions. Sort the resulting node lists lexicographically to select first, middle and last regions for the mechanism comparisons. Free-flow weights are t^0 and congestion weights are the TSTT marginal weight minus t^0, at the same total-flow state for both study and background responses.

B.7. Learning objectives and exact endpoint targets

This identity suggests learning the reference's weighted demand response where it affects spatial accounts. With training-only positive scales a_j, the boundary term is

$$\mathcal{L} = \mathcal{L}_{\text{flow}} + \lambda_J \mathcal{L}_{\text{response}} + \lambda_{\text{cut}} \mathcal{L}_{\text{boundary}}, \qquad \mathcal{L}_{\text{boundary}} = \frac{1}{NK} \sum_{q,j} \left[\frac{\mathcal{K}_j(h_j(q))}{a_j} \right]^2. \qquad \text{(B.5)}$$

The reference Jacobian is obtained from the existing implicit SUE sensitivity system (Appendix B.2). We compare ordinary flow supervision, flow plus response supervision, and the full objective. Derivative supervision itself is established in Sobolev training (Czarnecki et al., 2017), including finite-difference derivative targets (Yuan et al., 2022); the network-derived target here selects the part of weighted response error that changes boundary accounts. A circulation in $\ker B$ is invisible to $\mathcal{K}_j$, so flow and response losses remain necessary for the other attribution objectives. The target matches the reference's actual response; it does not force its true internal attribution divergence to zero or replace TSTT weights with uniform weights. Equation (16) is a path inequality; finite training-point losses alone do not certify an entire demand domain.

For the following finite-change statement, the outcome is explicitly separable:

$$g(F) = (g_e(F_e))_{e \in E}, \qquad T(F) = \sum_{e \in E} g_e(F_e).$$

Proposition B.1 (exact finite-change boundary target)**.** *Let f^L, f^S and the edgewise outcome g be continuously differentiable on an open demand domain, and let $q^{\pm} = q \pm he_j$ with their entire segment in that domain. Put*

$$z_j^r = g(f^r(q^+)) - g(f^r(q^-)), \qquad r \in \{L, S\}.$$

For the attribution of this single-OD change,

$$\max_{U: s_j \in U,\ t_j \notin U} |\phi_j^L(U) - \phi_j^S(U)| = \mathcal{K}_j(z_j^L - z_j^S). \tag{B.6}$$

If all sufficiently small single-coordinate changes have matching accounts on every separating boundary at every point of the domain, then $BD^L J_j^L = BD^S J_j^S$ throughout the domain. Integrating yields matching boundary attribution columns for any joint demand path in that domain (with the same columnwise path convention).

Appendix A.7 proves both statements. The first is an exact finite-change target, whereas the pointwise response objective optimizes the upper bound in Eq. (16); integrating before taking $\mathcal{K}_j$ can retain cancellations. The second statement requires a domain-wide local premise and is not a guarantee from finite training data. With several ODs changing, the endpoint outcome difference only identifies the sum of their attribution columns.

For sampled single-OD endpoint pairs, a direct training objective is

$$\mathcal{L}_{\text{pair}} = \mathcal{L}_{\text{flow}} + \lambda_{\text{edge}} \frac{1}{N} \sum_{(q,h,j)} \frac{\|z_j^L - z_j^S\|_2^2}{b_j^2} + \lambda_{\text{cut}} \frac{1}{N} \sum_{(q,h,j)} \left[\frac{\mathcal{K}_j(z_j^L - z_j^S)}{a_j} \right]^2, \tag{B.7}$$

with positive training-only scales a_j, b_j. The edge term controls changes invisible to the boundary seminorm, whose kernel is $\ker B$. Two reference endpoint equilibria supply z_j^S; training uses ordinary parameter backpropagation through two predictions, without reference Jacobian labels or input-Jacobian differentiation. Actual label-generation and training costs must still be measured.

B.8. Finite-change supervision: protocol and additional results

All four methods continue each of the same five pretrained conservative models for 1,200 Adam updates (learning rate 3×10^{-5}, flow batch 32, query batch 8). The flow data and random minibatch sequences are shared. Each query has 64 training and 16 validation endpoint pairs. Pair centers are sampled independently, coordinatewise, from $[0.65, 1.3]$ times baseline demand; half-widths are 2.5% and 15% of the queried baseline demand, equally represented. Only the queried demand changes within a pair; all background route shares remain responsive. Edge and boundary variants share exactly the same endpoint labels. Their weights are fixed at 1 and 0.01, respectively, without a new weight search. Training-only root-mean-square scales are used for each query.

All methods use the same 96 validation pairs every 100 updates. Among checkpoints whose mean relative flow error is at most 1.1 times the initial checkpoint's error, the minimum normalized boundary mean-square error is selected. If none qualifies, the minimum-flow-error checkpoint is retained. This fallback occurs for seeds 0, 2, and 3 of both endpoint variants; the flow threshold is therefore a selection preference, not a guarantee satisfied by every selected model. Test paths never enter selection.

For each path–query pair, let $e_j = \widetilde{C}_j^L - \widetilde{C}_j^S$ and $c_j = \widetilde{C}_j^S$. Column, system and boundary relative errors are respectively $\|e_j\|_2 / \max(\|c_j\|_2, 10^{-6})$, $|\mathbf{1}^\top e_j| / \max(|\mathbf{1}^\top c_j|, 10^{-6})$, and $\mathcal{K}_j(e_j) / \max(\mathcal{K}_j(c_j), 10^{-6})$. The absolute boundary error is $\mathcal{K}_j(e_j)$; geometric error replaces $\mathcal{K}_j$ in numerator and denominator with the maximum absolute account over the prescribed screenline family. Percentages multiply these ratios by 100.

The 24 prespecified test paths comprise six paths each with one, two, three, or six active queries. Cyclic combinations balance query participation within each group; paths include same-direction and mixed-direction changes of 15–30% of baseline demand. Each path has 41 states. The flow column is the five-seed mean relative error on the original held-out demand-state split, not a new-path statistic. Other main-table statistics first average active queries and seeds within each path, then summarize the 24 paths; seeds do not increase the number of independent demand paths. The absolute lower bound on relative-error denominators

is 10^{-6}; the smallest observed reference boundary scale is 92.55. The largest reference completeness residual is 3.38×10^{-8}, and the largest learned-model residual is 2.98×10^{-6}. The single-OD fixed-budget comparison instead uses exact endpoint columns and no quadrature.

Additional path-median straight-screenline errors for flow, endpoint-edge, endpoint-edge-plus-boundary, and response supervision are respectively 7.91%, 7.40%, 7.97%, and 5.81%; flow-increment errors are 4.81%, 4.89%, 5.09%, and 3.55%. Across the five seeds, the ranges of mean boundary errors, giving equal weight to the 24 paths, are respectively $[13.36, 24.12]$%, $[12.29, 17.69]$%, $[12.27, 20.58]$%, and $[10.53, 17.46]$%.

The 960 new endpoint solves take 64.0 s. The response implementation additionally solves all 480 pair centers and computes their sensitivities in 53.1 s; 384 center derivatives enter training, while the 96 validation-center derivatives are unused but included in this measured cost. Test reference generation takes 74.4 s. These figures exclude the existing flow dataset and pretrained checkpoints. Median continuation times per run, including initialization, validation, and model saving, are 246.0, 422.2, 275.1, and 297.1 s in the above method order. Five seed workers run concurrently, with methods executed sequentially within each worker; total continuation wall time is 1,303.5 s. Shared-machine contention and stage overlap prevent interpreting these per-method wall times as an algorithmic speed comparison. The corresponding median times for attribution and evaluation of all 24 paths, including forward passes, six directional derivatives, integration, and metric assembly, are 15.9, 15.8, 13.8, and 11.0 s. Equal update counts do not imply equal reference-data or computational costs.

C. Numerical accuracy and supporting results

C.1. Grid refinement, derivative, and solver checks

All differences here are observed numerical comparisons, not certified bounds. Direct attribution-column differences are at most 0.178% for 11 versus 41 states and 1.1×10^{-4}% for six-query/single-OD 41 versus 81 states. Near-zero accounts on 41/81/161 grids change by less than 5×10^{-6} percentage points (absolute < 0.01); the learned 201/801-state relative difference is 4.97×10^{-6}. Implicit versus re-solved central-difference Jacobian errors are at most 7.6×10^{-5} (halved step 2.5×10^{-5}), or 3.0×10^{-8} for six-query/single-OD columns. Newton versus successive-averages differences over 1,353 solves are at most 1.31×10^{-11}. Maximum fixed-point and balance residuals are 3.10×10^{-8} and 1.82×10^{-11} veh/h. Scalar completeness relative discrepancies are at most 3.36×10^{-5} for frozen/re-routing settings and 1.9×10^{-7} for single-OD checks. Per-edge endpoint checks give relative column errors 7.7×10^{-8}–1.1×10^{-6} and maximum entry residuals 5.4×10^{-4}–1.6×10^{-2} TSTT units. Per-query records remain in the arrays. The direct $(10, 16)$ account change is 24.80% on its 41-state grid; Figure 2 uses the seed-zero learned model on 201 states (24.8%).

C.2. Normalization, cancellation, and sensitivity

Error-metric definitions. The surrogate test flow error is, within each seed, the mean over the 857 test demands of $\|f(q) - F^{\mathrm{SUE}}(q)\|_2/\|F^{\mathrm{SUE}}(q)\|_2$, reported as the median over the five seeds; Jacobian column errors $\|J_j^{\mathrm{learned}} - J_j^{\mathrm{SUE}}\|_2/\|J_j^{\mathrm{SUE}}\|_2$ are pooled over the $5\times41\times8$ seed–point–OD columns; attribution-column errors use the per-column norm over the 40 seed–OD columns on the common 41-point grid.

Same-path flow, sensitivity, and boundary-account errors. The test-split flow error is not paired with the Jacobian comparison; re-summarized on the same 41 zero-background quadrature points for each seed (median / 95th percentile / maximum, percent): flow error 0.136 / 0.682 / 1.331; Jacobian column error 0.516 / 3.903 / 9.562 (maximum at seed 3, OD $(16, 10)$, $\lambda = 0.925$; $5\times41\times8$ seed–point–OD units); attribution-column errors median 0.349%, maximum 1.80%; and E_j^{cut}, which evaluates both models' accounts at every connected boundary of a query's family on the same path (the unrestricted-family counterpart is virtually identical), at 0.204 / 0.650 / 1.739 (40 seed–OD units; largest connected error at OD $(16, 10)$, seed 3).

Table C.1 supplies the per-query absolute origin-cut denominators for Table 4. The setting-C link sums C_j span 2.078–4.282×10^5 TSTT units, exceeding origin-cut accounts by factors of 1.01–2.37; their sum matches the endpoint TSTT change within 2.3×10^{-5} relative.

The node breakdown in Table C.2 complements the fixed boundary comparison of Section 6.3.

For the learned zero-background low-range group, five-seed median pointwise imbalance budgets $100A_j/|\phi_j(\{s_j\})|$ lie between 0.22% and 0.95%, versus 26.85–59.71% for the high group; the low-group

Table C.1
Absolute direct-SUE origin-cut accounts $|\phi_j(\{s_j\})|$, in TSTT units, for the eight queries in Table 4. A, B, C use their respective 41-point columns; entries rounded to one decimal.

OD	A	B	C
$(17, 10)$	29,372.1	249,729.2	224,954.3
$(10, 17)$	46,977.2	317,861.9	245,178.7
$(16, 10)$	41,994.3	352,858.5	180,671.5
$(10, 16)$	57,468.9	405,374.2	272,743.2
$(15, 10)$	33,859.4	226,878.2	201,351.2
$(10, 15)$	34,056.4	258,682.3	232,160.6
$(10, 11)$	28,807.1	339,651.4	202,928.2
$(11, 10)$	28,010.4	324,427.0	204,802.3

Table C.2
Signed node divergence in the re-routing setting, at the same joint equilibrium state and total-flow gradient at each of the 41 quadrature points (study includes all eight query ODs; TSTT units of Appendix B.1; entries rounded independently, so a component sum may differ from the rounded total in its last digit).

OD	Node	Study	Background	Total
$(10, 16)$	15	-973.1	$49,103.4$	$48,130.3$
$(10, 16)$	8	9.6	$42,102.7$	$42,112.3$
$(10, 16)$	20	0.2	$41,805.4$	$41,805.6$
$(10, 11)$	5	0.5	$35,952.1$	$35,952.6$
$(10, 11)$	12	0.0	$-23,858.1$	$-23,858.1$
$(10, 11)$	9	-0.2	$-18,557.6$	$-18,557.7$

median cancellation factors are at most 0.313 (one seed reaches $\kappa = 0.852$), so small pointwise imbalance mainly limits the low-group range, while the high group's median κ is below 8×10^{-5}.

Zero-background route-choice dispersion checks at $\theta \in \{0.5, 1, 2\}$ (eleven-point direct-SUE calculations) retain the large/small separation — large ranges 27–64%, the others below 9.4%.

C.3. Full-background six-query surrogate validation

The six-query conservative surrogate predicts shares over all 5,280 paths, with the architecture of Appendix B.3, dataset seed 20261025 and batch size 256. Its 4,214 target solves have maximum path residual 3.5×10^{-8}; learned conservation residual is at most 3.6×10^{-11}. Held-out flow error is 0.040% (seed median; range 0.035–0.056%). On the 41 common path states, flow errors are 0.037–0.065%, but increment errors are 5.0–7.0% against a reference increment norm 2,982.7 (2.7% of flow level). Median column errors are 81.2% for $(7, 18)$, 31.1% for $(4, 5)$ and 7.4–11.8% for the others. System contribution errors reach 14.7% and 10.3% for those two queries, versus at most 3.7% for the sign-crossing queries. Table C.6 retains extrema and boundary errors; learned column completeness is within 6×10^{-6} relative.

An exact two-term split of (2) (L/S: learned/direct) identifies where this error arises:

$$\widetilde{C}^{L}_{:,j} - \widetilde{C}^{S}_{:,j} = \Delta q_j \int_0^1 (D^L_\lambda - D^S_\lambda) J^S_j \, \mathrm{d}\lambda + \Delta q_j \int_0^1 D^L_\lambda (J^L_j - J^S_j) \, \mathrm{d}\lambda, \tag{C.1}$$

satisfied entrywise within 10^{-10} TSTT units. The weight-shift term has querywise median norm at most 0.2% of the reference column; the response term is larger in every one of the thirty seed–query units. Their norm sum is at most 1.04 times the realized error norm. Signed scalar terms can cancel, so these norms are not additive shares. Small-band endpoint columns have 4.8–15.9% error and system values within 2.7% of reference, while every seed retains the reported signs.

C.4. Sensitivity to candidate paths and route-choice dispersion

The path-set comparison expands all 528 OD sets to twenty candidates each (the first ten unchanged), with demands, BPR functions, $\theta = 1$, and 41 quadrature points fixed: across the eight queries, absolute unrestricted and connected ranges change by at most 0.977% and 1.011%, with the three largest internal-node terms keeping their ordered node IDs (moving at most 1.43%). The $\theta \in \{0.5, 2\}$ check splits the specified $(10, 16)$ boundary difference into study/background components $-7{,}404.9/-2{,}371.5$ at $\theta = 0.5$ and $+115.3/+235.7$ at $\theta = 2$; for all three additional models, implicit derivative columns match re-solved half-step central differences at the demand-path midpoint (maximum 0.00249%, completeness gap 2.85×10^{-5}).

C.5. Six-query interval, mechanism, boundary-family, and scenario checks

Extremal ranges and negative shares. Unrestricted extrema follow the closed form (11), connected extrema the same enumeration as the connected ranges (Table C.3); the negative shares in Section 6.4 are exact counts over the connected families (4,621–16,943 admissible sets per query), and all displayed values come from the 41-point columns.

Table C.3
Minimum–maximum boundary-account ranges for the three sign-crossing queries of Table 5, in TSTT units (Appendix B.1), rounded to integers. Baseline: unrestricted extrema (closed form (11); connected extrema in Table 5); next four rows: connected ranges under the model-setting changes and the single-OD $0.2 \to 1.6$ re-run (unrestricted intervals within 1% of the baseline row); last two: small-interval bands, with C_j positive ($+7{,}202$ to $+33{,}734$ across the three queries) and the control $(6, 8)$ positive ([5,079, 9,623] and [10,157, 19,246]; single-OD [35,352, 66,861]).

Range	$(1, 19)$	$(1, 15)$	$(7, 12)$
Baseline (10 paths, $\theta = 1$): unrestricted	[−19,116, 26,696]	[−16,240, 44,137]	[−45,241, 61,650]
Twenty paths, $\theta = 1$: connected	[−13,451, 24,276]	[−8,329, 40,644]	[−17,728, 59,452]
Ten paths, $\theta = 0.5$: connected	[−13,024, 24,064]	[−8,107, 39,933]	[−17,633, 60,476]
Ten paths, $\theta = 2$: connected	[−13,016, 24,411]	[−8,233, 40,892]	[−19,587, 61,647]
Single-OD $0.2 \to 1.6$: connected	[−13,608, 24,414]	[−8,359, 40,745]	[−17,580, 60,006]
Single-OD ±10% band ($0.9 \to 1.1$): connected	[−1,947, 3,494]	[−1,201, 5,838]	[−2,517, 8,608]
Single-OD ±20% band ($0.8 \to 1.2$): connected	[−3,895, 6,988]	[−2,402, 11,677]	[−5,035, 17,215]

Connected minimum and divergence decomposition. Table C.4 reports, for each sign-crossing query, the connected minimizing boundary U_- at the baseline setting: its account, and the interior divergence sum $\phi_j(\{s_j\}) - \phi_j(U_-)$, decomposed as the table caption specifies, at the same joint state and gradient as the reported columns (additivity $\leq 1.2 \times 10^{-10}$). By Eq. (12), the negative divergence captured inside U_- (4,664.9, 6,960.1, and 19,811.5 for the three queries) covers 83%, 88%, and 72% of the gaps $\phi_j(U_-) - \phi_j^{\min}$, while the captured positive divergence (19,938.0, 18,194.2, 46,901.4) is 95.5%, 94.8%, and 85.7% of the query totals. The sign change arises differently across queries: study response dominates for $(1, 19)$ (largest node 15: own $+14{,}464.0$), study- and background-dominated nodes share the sign for $(1, 15)$, and both contribute about equally for $(7, 12)$ (node 8). The extremal node sets (one-based node IDs) are $U_-(1, 19) = \{1, 2, 6, 7, 8, 9, 15, 16, 18, 20, 21, 22, 24\}$ and $U_+(1, 19) = \{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 22, 23\}$; $U_-(1, 15) = \{1, 2, 6, 7, 8, 9, 16, 17, 18, 19, 20, 21, 24\}$ and $U_+(1, 15) = \{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 22, 23\}$; $U_-(7, 12) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 17, 19, 23\}$ and $U_+(7, 12) = \{7, 8, 9, 15, 16, 18, 20, 21, 22, 24\}$.

Table C.4
Connected minimizing boundary U_- at the baseline setting for the three sign-crossing queries, in TSTT units. Interior is the divergence sum over $U_- \setminus \{s_j\}$, decomposed into the query's own response, the other five query ODs, and the background response; node sets are the one-based lists in the preceding paragraph. Entries are rounded independently.

OD	$\phi_j(\{s_j\})$	$\phi_j(U_-)$	Interior	Own OD	Other query	Background
$(1, 19)$	1,771.6	−13,501.6	15,273.1	19,480.9	−59.4	−4,148.3
$(1, 15)$	2,948.1	−8,286.0	11,234.1	7,856.9	−26.7	3,404.0
$(7, 12)$	9,473.8	−17,616.2	27,089.9	14,002.3	−3.2	13,090.8

Neighbourhood families. Use undirected hop distance or shortest free-flow distance from the origin (minimum weight over parallel arcs). Include nodes at each attained radius, stop before including the destination, and retain only two-sided connected regions. Table C.5 reports all family ranges; all are positive with minima at the origin cut. For short-path queries $(7, 18)$, $(4, 5)$ and $(6, 8)$, only that cut is admissible. The thirteen-node free-flow ball at radius 16 for $(1, 19)$ is $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 24\}$, with account +10,756.0.

Table C.5
Origin-centred neighbourhood accounts, baseline six-query setting C (TSTT units). Counts are distinct admissible node sets within each family; sets shared by the two families are counted in each row.

OD	Neighbourhood	Count	Minimum	Maximum
$(1, 19)$	Hop distance	6	$1,771.6$	$15,803.5$
$(1, 19)$	Free-flow time	13	$1,771.6$	$15,803.5$
$(1, 15)$	Hop distance	5	$2,948.1$	$26,602.0$
$(1, 15)$	Free-flow time	14	$2,948.1$	$26,602.0$
$(7, 12)$	Hop distance	4	$9,473.8$	$31,524.1$
$(7, 12)$	Free-flow time	9	$9,473.8$	$32,139.4$

Straight screenlines. Coordinates use an equirectangular frame (longitude scaled by the cosine of mean latitude). Section 5.2's 36-direction construction, with duplicate regions removed, gives 35, 152, 32, 123, 23 and 148 regions in Table 5 order. All accounts agree with direct incidence aggregation below 10^{-9}. Refining directions from 5° to 1° leaves both negative minima unchanged; negative shares change from 1.3% to 1.8% for $(1, 19)$ and 6.8% to 6.2% for $(7, 12)$. Other queries stay positive; only $(7, 18)$'s minimum changes, from 642.4 to 641.4.

Model-setting checks. Re-solving the protocol of Appendix B.5 under twenty candidate paths per OD and $\theta \in \{0.5, 2\}$ at ten paths gives the ranges of Table C.3 (C_j within 3.9%); solver and finite-difference checks match Appendix B.5 levels.

Single-OD demand changes. Each single-OD demand path ($0.2 \to 1.6$ of published demand) re-solves the setting-C protocol with one moving coordinate. For the three crossing queries the connected ranges (Table C.3, single-OD row) still straddle zero, their extrema attained by exactly the joint-scenario sets U_- and U_+ (changes at most 0.9% and 0.5%; unrestricted intervals within 1% of the joint baseline) and the contributions stay positive (C_j = 50,322.2, 77,672.6, 117,476.8); the control $(6, 8)$ remains sign-consistent (C_j = 74,176.1, connected range in the caption). The small-interval bands (Table C.3, last two rows) follow the same protocol, the 41-point check ($(1, 19)$, $0.8 \to 1.2$) reproducing the endpoint column to 1.3×10^{-5} TSTT units per edge.

Endpoint identity checks. Appendix C.1 reports the per-edge endpoint-identity errors and their relation to completeness gaps on these same demand paths.

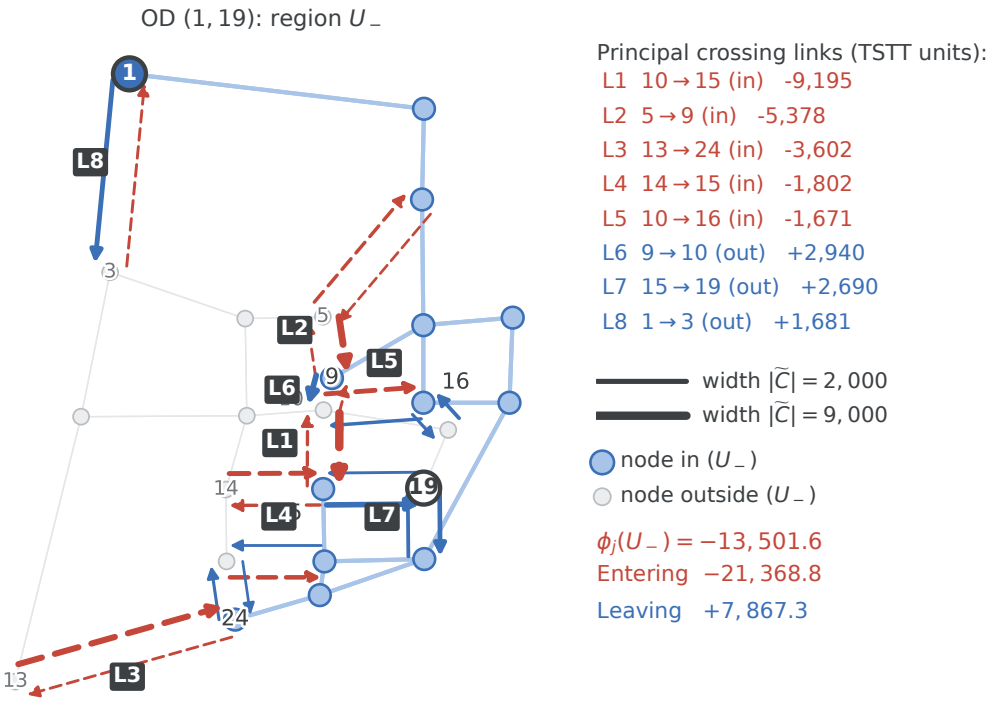


Figure C.1: The connected minimizing region U_- for OD $(1, 19)$ (setting C, six-query protocol). Shaded nodes form U_- (origin node 1 ringed; destination 19 outside). Crossing-link arrows keep the link direction; width scales with $|\widetilde{C}_{e,j}|$, and colour and style encode the signed contribution to $\phi_j(U_-)$, $a_{e,j} = \sigma_e \widetilde{C}_{e,j}$ ($\sigma_e = +1$ leaving, -1 entering; solid blue positive, dashed red negative), so a positive attribution on an entering link appears red. The side panel lists, in TSTT units, the five largest negative and three largest positive contributions among the 26 crossing links (L1–L5 negative, L6–L8 positive, "(in)"/"(out)" marking entering and leaving links), with the width scale and the membership legend.

C.6. Baseline surrogate envelopes

Table C.6
Full-background surrogate validation on the identical 41-state path of the six-query protocol (setting C; TSTT units; contributions and extrema rounded to integers, errors to one decimal). C_j^{learned} and the connected extrema: across-seed envelopes of per-model values (direct-SUE values in Table 5). Last two columns: five-seed *median* of the largest connected-boundary disagreement ϵ_j^{cut}, in absolute units and as a percentage of the direct connected span.

		Connected extrema: across-seed ranges			
OD	C_j: across-seed envelope	per-model minima	per-model maxima	ϵ^{cut} (med.)	ϵ^{cut}/span (med., %)
$(7, 18)$	[574, 681]	[335, 521]	[907, 1,299]	397.5	522.5
$(1, 19)$	[49,929, 51,452]	[−13,911, −13,334]	[23,683, 23,948]	1,567.4	4.2
$(4, 5)$	[2,535, 2,934]	[1,654, 2,148]	[2,946, 3,546]	1,050.8	99.5
$(1, 15)$	[77,416, 80,233]	[−7,717, −6,766]	[39,473, 40,076]	2,187.3	4.5
$(6, 8)$	[73,565, 75,224]	[36,412, 37,631]	[66,892, 67,762]	2,163.2	6.9
$(7, 12)$	[116,562, 118,706]	[−17,084, −15,198]	[59,965, 61,169]	4,111.3	5.3

C.7. Fixed-screenline surrogate comparisons

Sign preservation. All five baseline seeds retain the three connected-family sign crossings and positive system contributions. At the fixed reference extrema, $m_j = \min\{-\phi_j^S(U_-), \phi_j^S(U_+)\} - \epsilon_j^{\text{conn}}$ is positive,

Table C.7
Fixed straight screenlines: reference minimum and the five frozen flow-supervised surrogates at that same screenline (TSTT units). The last column gives the largest error over the screenline family, as an across-seed range.

OD	Reference minimum	Learned account: seed range	Maximum screenline error: seed range
$(7, 18)$	642.4	$[576.9,\ 789.8]$	$[141.0,\ 269.4]$
$(1, 19)$	-2766.6	$[-3886.6,\ -3365.1]$	$[857.9,\ 1430.8]$
$(4, 5)$	2652.3	$[2301.7,\ 2918.8]$	$[438.0,\ 648.3]$
$(1, 15)$	511.1	$[732.8,\ 2068.2]$	$[1140.6,\ 2553.0]$
$(6, 8)$	45230.2	$[46431.5,\ 47218.3]$	$[1276.2,\ 1988.1]$
$(7, 12)$	-2967.3	$[-2810.4,\ -1916.6]$	$[1736.8,\ 2499.2]$

Table C.8
Fixed-budget endpoint boundary error (%) on six independent single-OD test paths. Seeds are averaged within each path before taking the path median.

Method	300 updates	600 updates	1,200 updates
Flow	4.40	4.13	3.97
Flow + endpoint edge	4.47	4.35	4.16
Flow + endpoint edge + cut	4.51	4.38	4.17
Flow + response	3.52	3.16	3.02

Table C.9
Mean unrestricted boundary error (%) by number of active queries. Each group contains six independent paths, each balanced across queries.

Method	One OD	Two ODs	Three ODs	Six ODs
Flow	14.34	18.24	19.71	19.80
Flow + endpoint edge	11.11	15.72	16.06	18.93
Flow + endpoint edge + cut	13.08	16.65	15.09	19.56
Flow + response	12.67	16.30	13.45	13.73

with smallest value 5519 TSTT units. This verifies sign preservation without implying accurate column magnitudes.

At the fixed direct-model minimizing screenline, every baseline seed retains the negative sign for both queries: $(1, 19)$ gives $[-3886.6, -3365.1]$ against the reference -2766.6, and $(7, 12)$ gives $[-2810.4, -1916.6]$ against -2967.3. Their largest family-wide errors across seeds are 1430.8 and 2499.2, below the respective reference sign margins. These are screenline-specific certificates; they supplement the separate connected-family result above (Appendix C.7).

The eight-query path-set and dispersion checks are collected in Appendix C.4. Boundary variation persists, although the specified $(10, 16)$ comparison $\{10\} \to \{10, 17\}$ changes sign across θ; these are different fixed-path models, not convergence to unrestricted SUE.

C.8. Final learning comparison: budgets and groups

Tables C.8 and C.9 report the fixed-budget and path-group breakdowns of the final comparison.

C.9. Input-stratified OD coverage

We select queries using only the published positive OD demand and directed free-flow shortest-path time. Each attribute is divided at its empirical tertiles, keeping ties in the lower group. In each of the nine cells, OD pairs are ordered lexicographically and the pairs at zero-based positions $\lfloor N/3 \rfloor$ and $\lfloor 2N/3 \rfloor$ are retained. Empty cells remain empty and a singleton is retained once. This yields 18 queries from 528 positive demands. For each query, only its demand moves from $(1 - a)q_j$ to $(1 + a)q_j$, $a \in \{0.1, 0.2\}$; all other demands remain at their published levels with endogenous route choice. Four endpoint equilibria per

Table C.10
All input-stratified queries. Entries are geometric-account spreads for $\pm 10\%$ and $\pm 20\%$ intervals and their ratios to the exact unrestricted spread. Absolute values use the Sioux Falls cost units. No query is removed based on its attribution.

OD	S^{10}_{geo}	S^{20}_{geo}	$S^{10}_{\mathrm{geo}}/S^{10}_{\mathrm{all}}$	$S^{20}_{\mathrm{geo}}/S^{20}_{\mathrm{all}}$
(4,3)	53.0	105.9	0.353	0.353
(12,3)	4.8	9.5	0.417	0.417
(6,1)	816.1	1632.1	0.651	0.651
(15,6)	2009.1	4018.1	0.672	0.672
(6,21)	1176.4	2352.8	0.668	0.668
(18,14)	889.5	1778.9	0.511	0.511
(8,4)	7235.2	14470.6	0.646	0.646
(19,8)	3266.6	6533.2	0.374	0.374
(11,6)	2434.9	4869.9	0.626	0.626
(16,2)	4366.2	8732.3	0.545	0.545
(8,21)	1730.7	3461.5	0.424	0.424
(14,8)	3318.4	6636.8	0.495	0.495
(11,15)	5054.3	10108.8	0.337	0.337
(17,9)	6509.7	13037.1	0.696	0.695
(10,12)	11713.3	23426.2	0.539	0.539
(13,10)	14211.4	28426.4	0.667	0.667
(1,10)	9336.8	18673.5	0.603	0.603
(10,1)	9428.5	18856.8	0.606	0.606

query give the exact single-OD edge attribution by Eq. (19), avoiding quadrature. The geometric family uses the same 36 pre-specified directions and connected-side filter as the screenline experiment. The median geometric-to-unrestricted spread ratios are 0.574 and 0.574 for the two widths. These results describe this fixed input-stratified sample; they are not a probability sample of all demand conditions. Endpoint differences identify total edge accounts, but do not identify the same-state study/background response decomposition. That decomposition is evaluated separately in the mechanism experiments.

C.10. Supplementary pointwise-supervision comparisons

Two earlier comparisons use the same five full-background flow-trained models, original data split and three-layer 128-unit tanh architecture with 5,280 path logits. All continuation runs use 600 Adam updates, learning rate 3×10^{-5}, flow batches of 32 and response batches of eight, cycling six queries. Reference derivatives at 256 training states use sampling seed 20261029; 64 additional derivative states are reserved for validation. Flow scales are 0.01 times edgewise training root-mean-square flows; response and boundary scales are querywise training root-mean-square column norms and $\mathcal{K}_j$ values, respectively. All losses average squared normalized errors. Model response derivatives are obtained by automatic differentiation.

The fixed-coefficient comparison sets both added loss coefficients to one, uses shared seed-specific flow minibatches, and retains the final update without selection. Four test paths, generated before training with seed 101029, have one, two, three and six active queries, mixed directions and baselines in $[0.65, 1.3]$ times published demands. Each has 41 quadrature states. Attribution metrics summarize 60 active query–path–seed units, flow metrics five seeds and increment metrics 20 path–seed pairs. Boundary errors divide by $\mathcal{K}_j(\widetilde{C}^S_{:,j})$; column, system and increment errors divide by the corresponding reference norm or absolute value. These descriptive units are not treated as independent paths for inferential statistics. Adding the boundary term increases pooled boundary error from 9.64% to 22.25%, although it improves 14/15 $(7, 18)$ units and 8/10 $(4, 5)$ units; the other query groups worsen. Reference labels and test paths require 83.3 s. Median continuation times are 105.6, 184.8 and 244.5 s for flow, response and boundary methods. Maximum conservation residual is 4.6×10^{-11}; relative quadrature completeness discrepancy is at most 5.1×10^{-6}.

The calibrated comparison selects λ_{cut} from $\{0.001, 0.01, 0.1\}$ on seed zero, using the 64 reserved validation states. It minimizes normalized boundary-response mean square subject to validation flow error at most 110% of the response comparator. The chosen coefficient is then applied to all five seeds, reusing its seed-zero fit: seven structural fits in total. Validation flow errors are 0.0610% for the response comparator and

Table C.11
Equal-update continuation from five flow-trained checkpoints on four independent demand paths. Entries are medians over 60 active query–path–seed combinations, except flow (five seeds, held-out split) and increment error (20 path–seed pairs). Boundary error uses all separating cuts and is normalized by the corresponding maximum absolute reference account.

Method	Flow (%)	Increment (%)	Column (%)	System (%)	Cut abs.	Cut (%)
Original	0.04	5.18	14.11	2.08	260.55	9.93
Flow continuation	0.03	5.34	12.29	1.48	268.81	9.98
Flow + response	0.03	5.48	11.92	1.65	283.64	9.64
Flow + response + boundary	0.12	14.07	32.01	1.97	708.35	22.25

Table C.12
Exploratory validation-calibrated continuation ($\lambda_{\mathrm{cut}} = 0.01$), tested on four new independent demand paths (seed 101030). Entries are medians over 60 active query–path–seed combinations, except flow (five seeds, held-out split) and increment error (20 path–seed pairs). Boundary error uses all separating cuts and is normalized by the corresponding maximum absolute reference account.

Method	Flow (%)	Increment (%)	Column (%)	System (%)	Cut abs.	Cut (%)
Original	0.04	5.40	12.22	1.65	289.36	7.47
Flow continuation	0.03	5.12	11.45	1.32	261.00	7.44
Flow + response	0.03	5.32	11.69	1.33	264.80	7.44
Calibrated boundary	0.03	5.64	11.92	1.68	275.14	7.74

0.0611%, 0.0631%, 0.0804% for the candidates; corresponding boundary-response mean squares are 0.7040, 0.6852, 0.6097, 0.4532. Thus 0.1 fails the flow criterion and 0.01 is selected. This additional search budget precludes an equal-search advantage claim. Four new test paths use seed 101030 and the same active-query templates, with evaluation after selection; the original paths remain diagnostic data. Calibration gives no overall improvement over response supervision: new-path boundary error is 7.74% versus 7.44%, and column error 11.92% versus 11.69%. For $(4, 5)$ all ten paired boundary errors improve (group medians 48.65% to 44.21%); for $(7, 18)$, eight of fifteen improve but the group median rises from 85.70% to 99.23%. On the original diagnostic paths, calibrated flow, increment, column, system and boundary errors are 0.0342%, 5.35%, 12.56%, 1.72%, and 324.11 units (10.09%). Candidate elapsed durations sum to 829.4 s (overlapping jobs, not CPU time); selected-fit median time is 211.8 s, new reference paths take 13.6 s, and the complete calibration/evaluation run takes 927.0 s. New-path maximum conservation residual and relative completeness discrepancy are 4.6×10^{-11} and 9.2×10^{-6}. Complete paired results remain in the numerical arrays. Neither supplementary comparison supports aggregate superiority of boundary supervision.